MANCHESTER 1824

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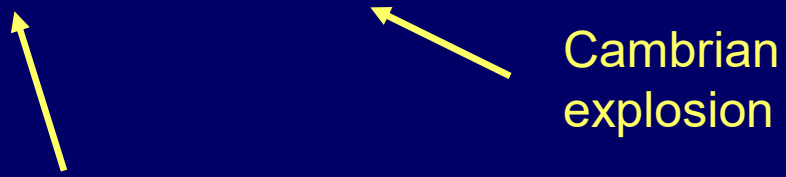
EART11100

How to Build a Planet

Class 7

The Proterozoic Earth: the Dawn of a Habitable World

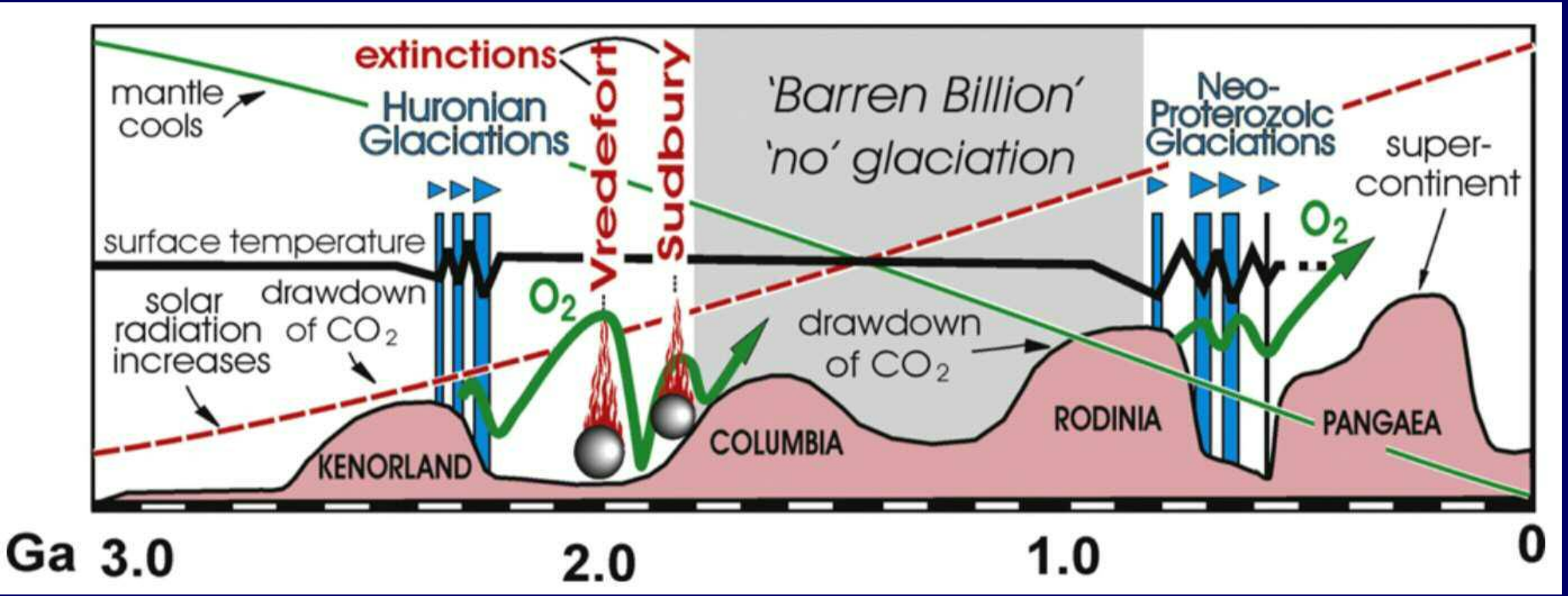
2420 Ma – 539 Ma



Attendance: 197137

Overview of the Proterozoic

Palaeoproterozoic Mesoproterozoic Neoproterozoic



Assembly and break up of supercontinents as mantle cooled and plate tectonic processes evolved

Luminosity of sun increased

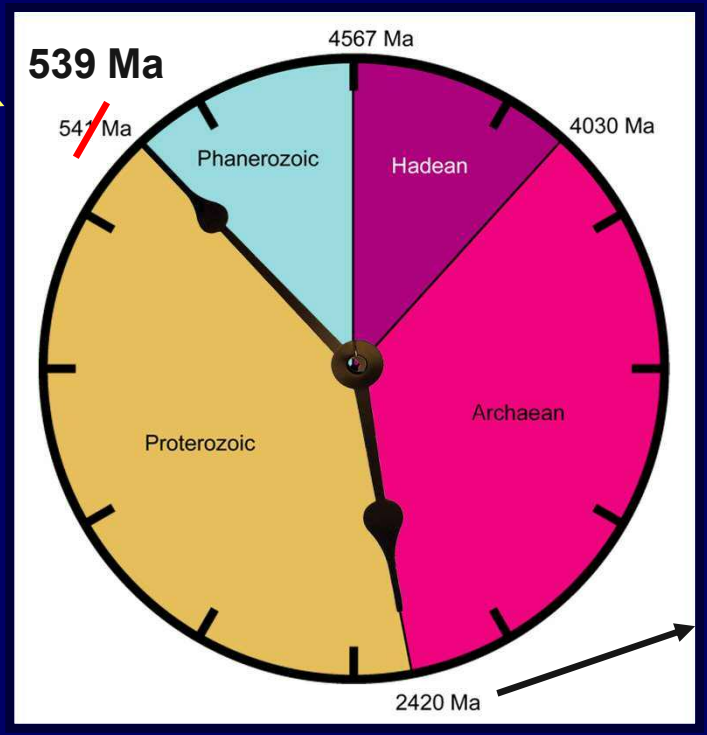
Rise (and fluctuations) of atmospheric oxygen levels due to photosynthesis and burial of organic carbon in sedimentary rocks

Oceans changed from reducing to aerobic as life evolved

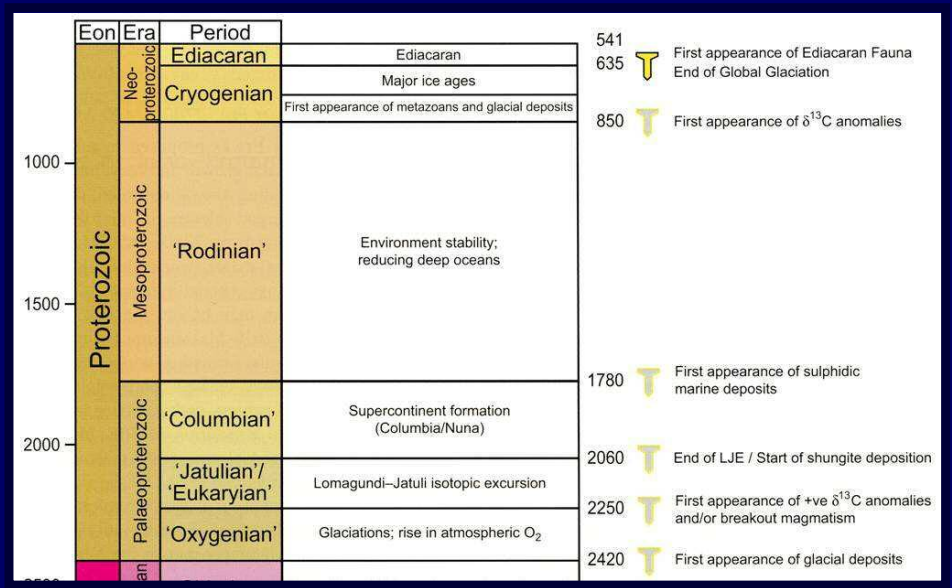
Major glaciations in Palaeoproterozoic and Neoproterozoic- linked to supercontinents and CO_2 (greenhouse gas) draw down from the atmosphere

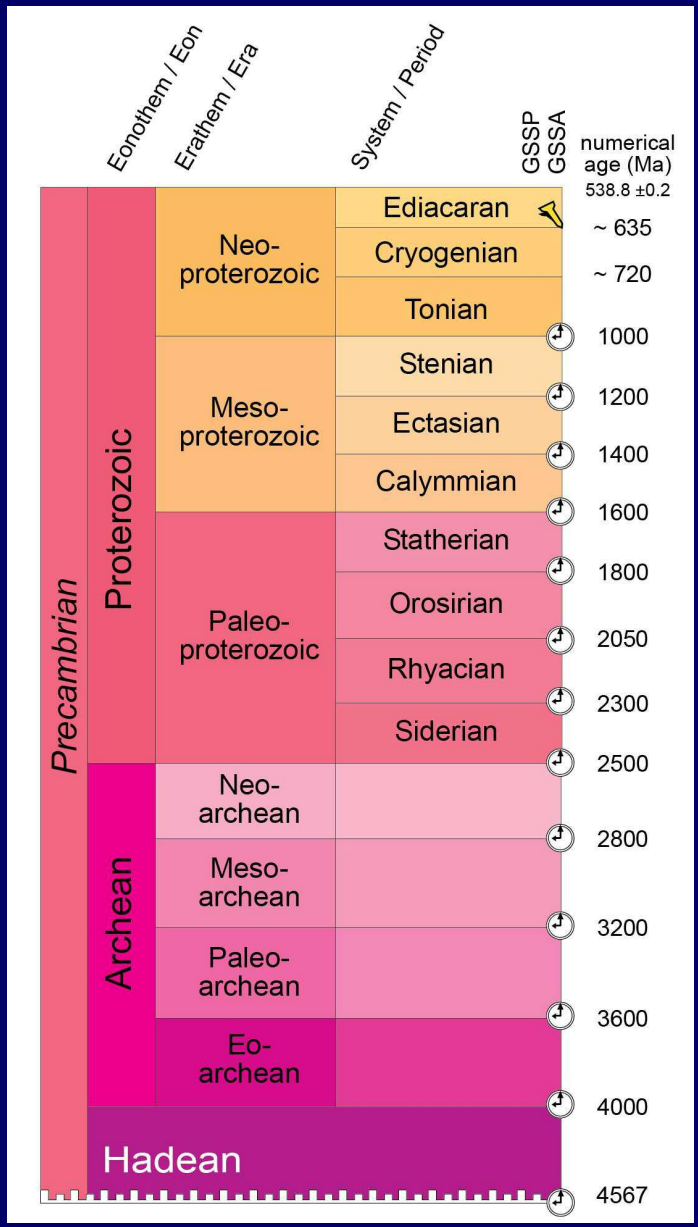
The Proterozoic Earth: Greek ‘earlier life’

Cambrian explosion



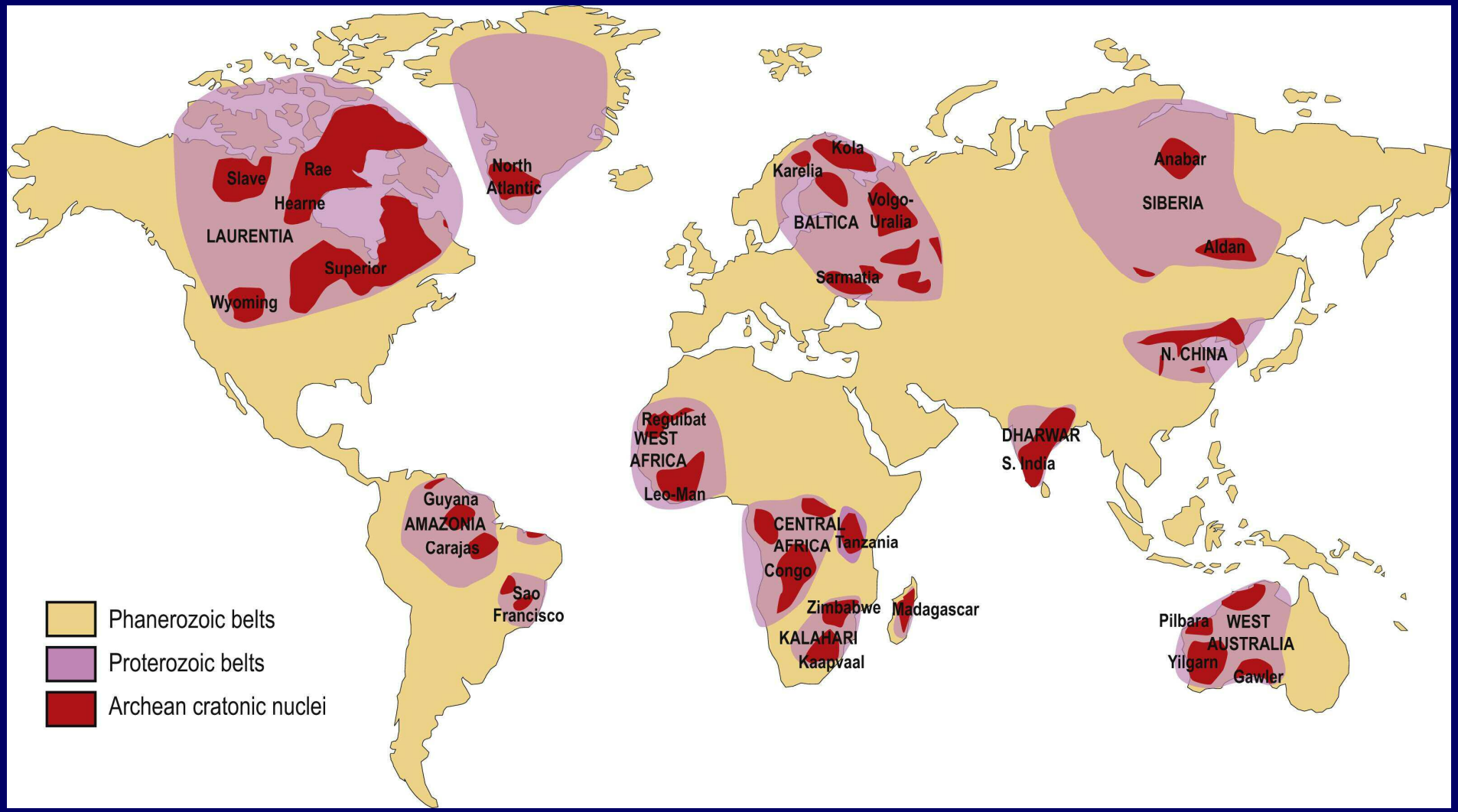
first globally distributed glacial deposits





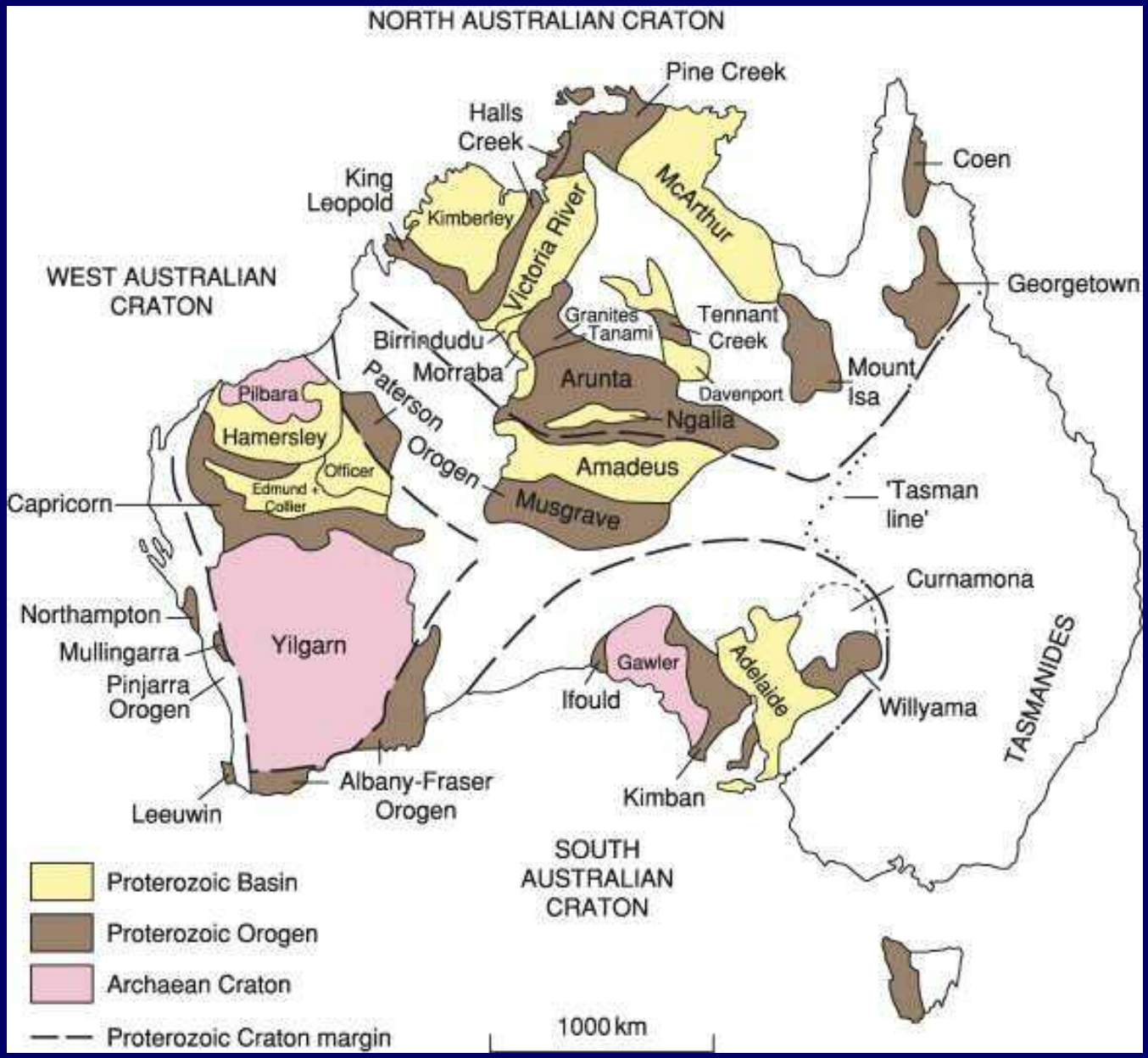
Note spelling: **Archaean** **Palaeo**

Age of continental cratons around the world today



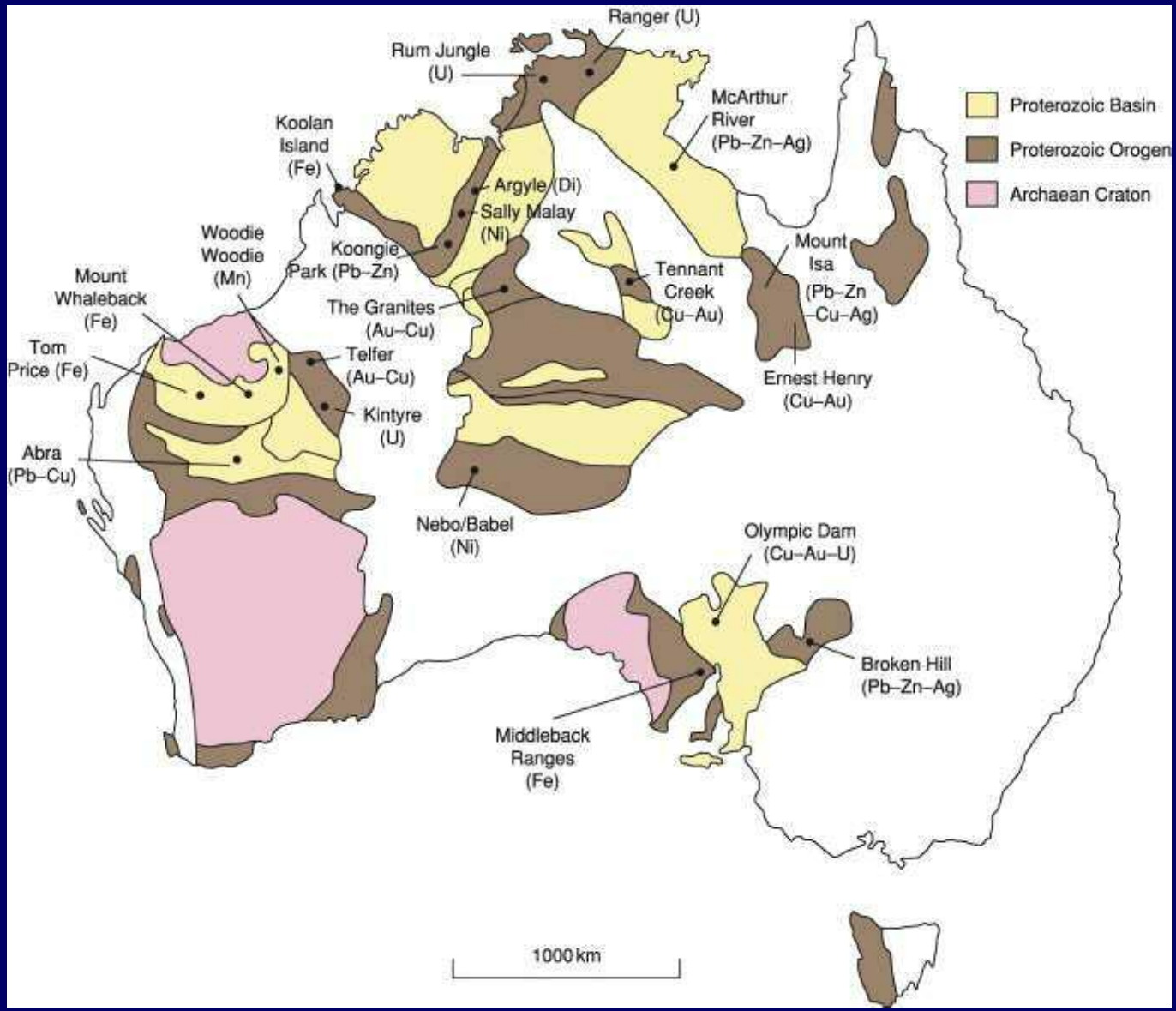
The global distribution of Proterozoic rocks and their association with Archaean cratons

Proterozoic orogens and sedimentary basins



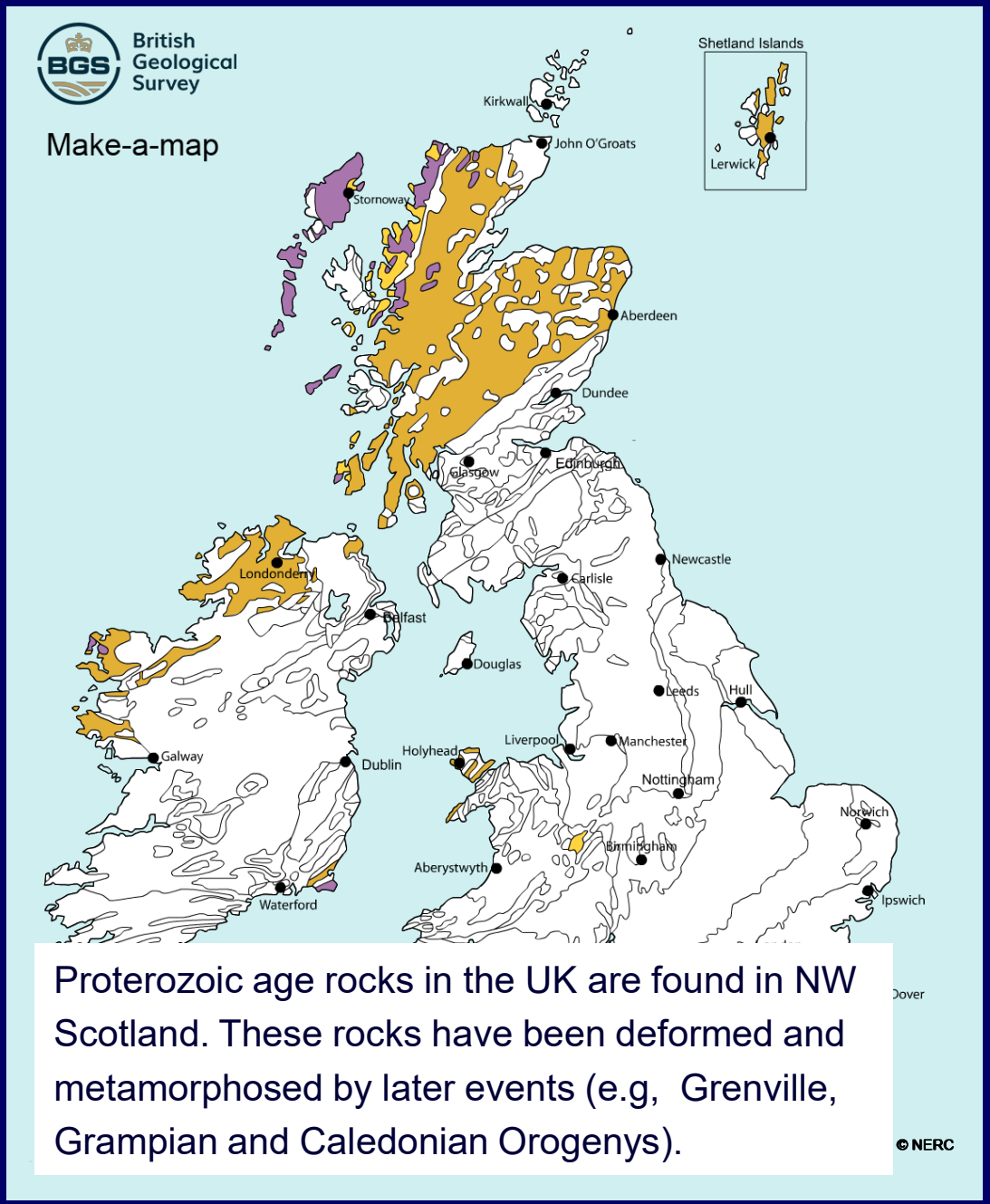
Distribution of Archaean and Proterozoic age rocks of Australia

Economic significance of Proterozoic rocks



Significant economic deposits in the Proterozoic rocks of Australia

Distribution of Proterozoic rocks in the UK

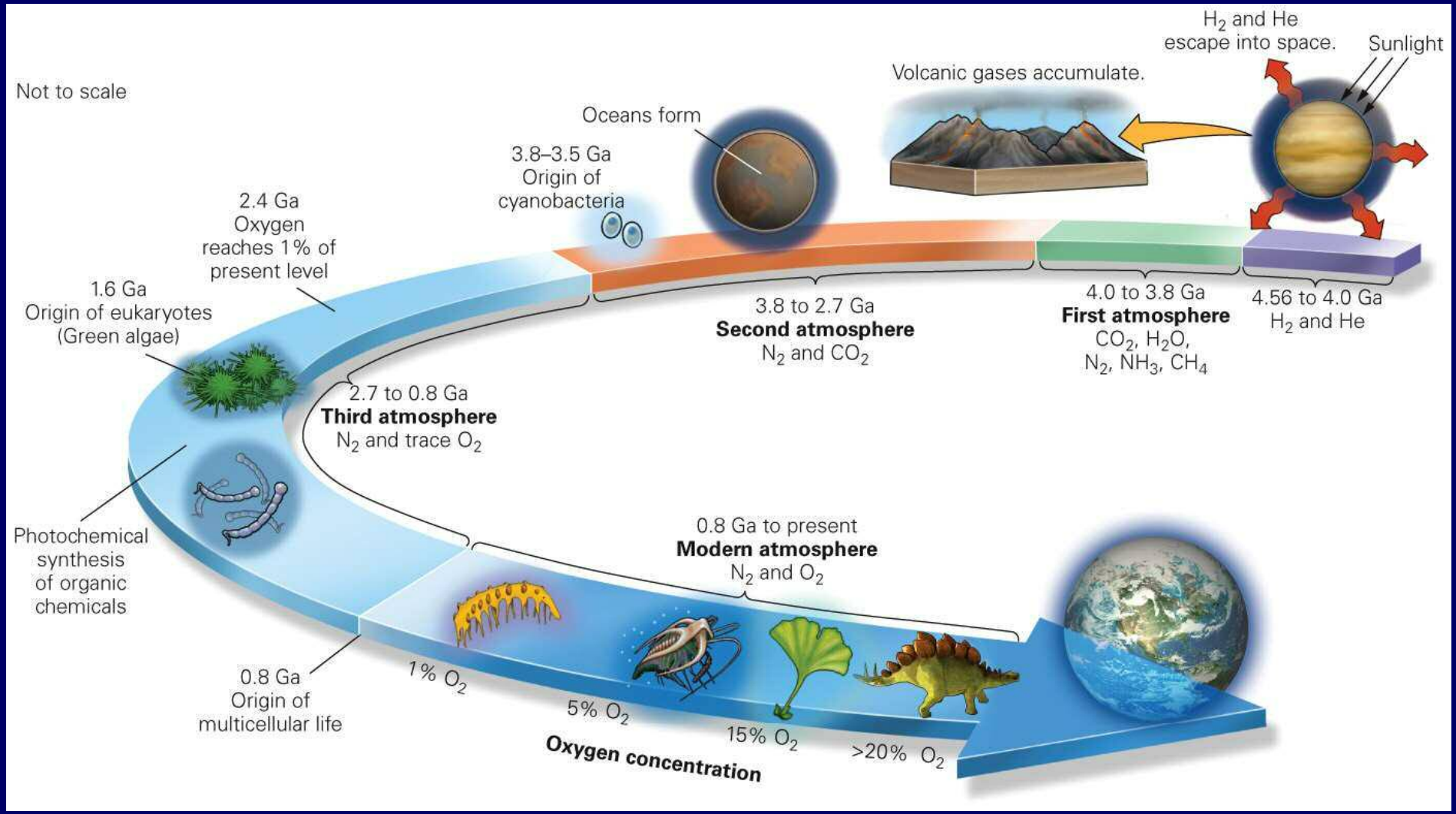


Proterozoic age rocks in the UK are found in NW Scotland. These rocks have been deformed and metamorphosed by later events (e.g, Grenville, Grampian and Caledonian Orogenys).

Scourie dykes (2418-2375 Ma) → rifting and break up of Archean supercontinent Kenorland

Port Askaig Tillite (Argyll Group) → Marinoan glacial event (645-635 Ma)

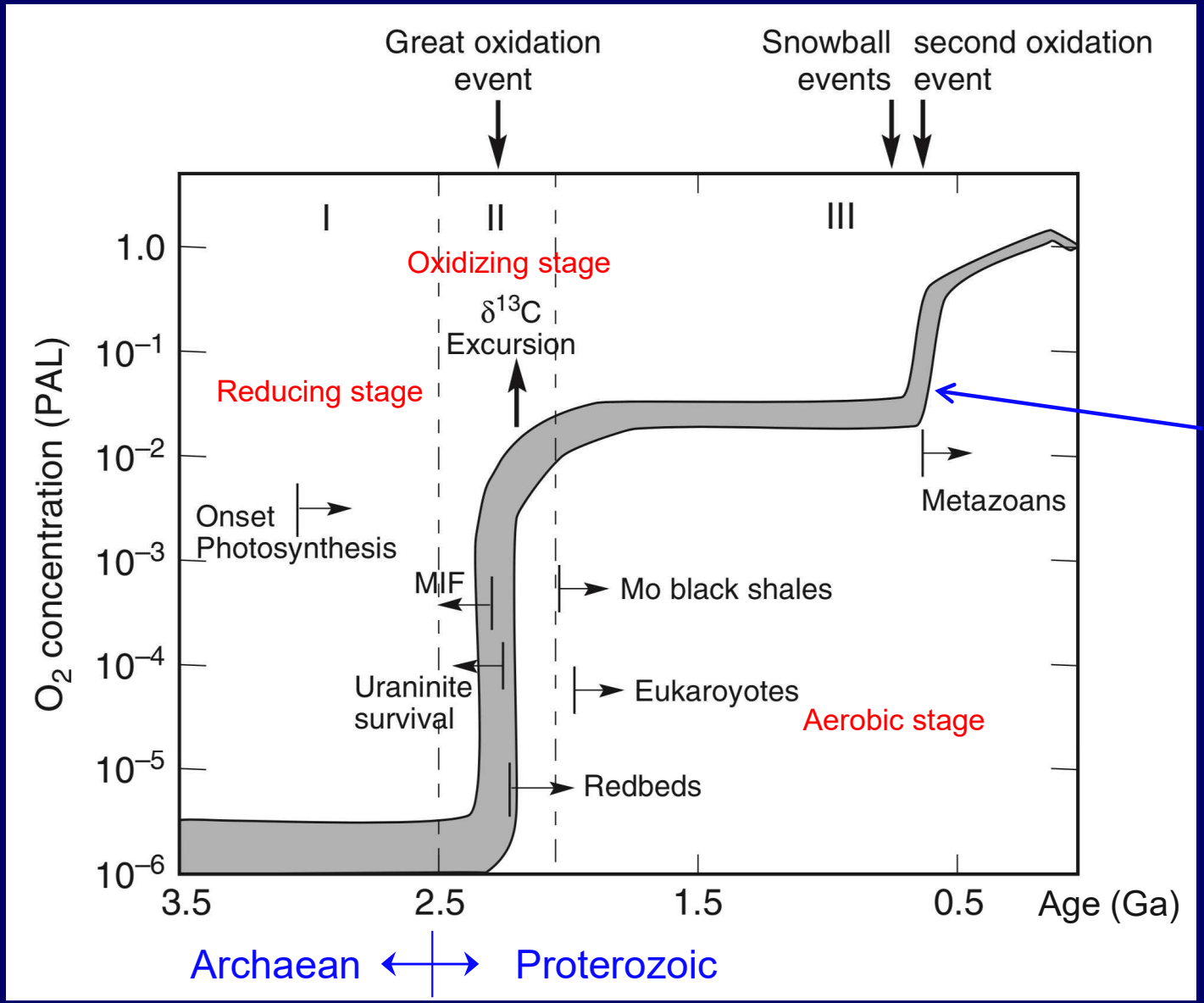
Key events of the Proterozoic: The oxygen revolution



Evolution of the Earth's atmosphere through geologic time

The most distinctive characteristic of Earth's atmosphere today is the presence of free oxygen

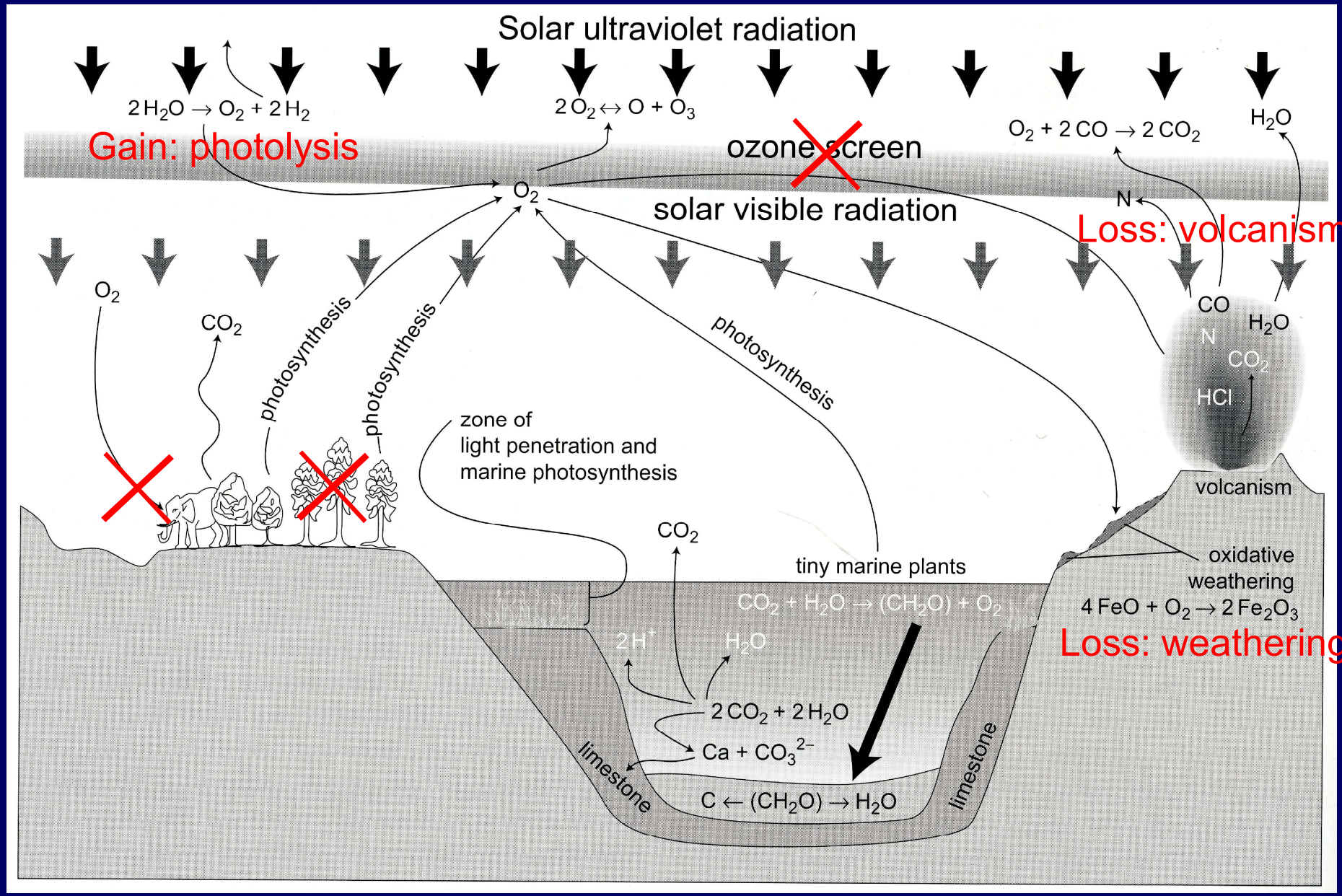
History of oxygen levels on Earth



The rise in atmospheric oxygen through the Proterozoic is not constant

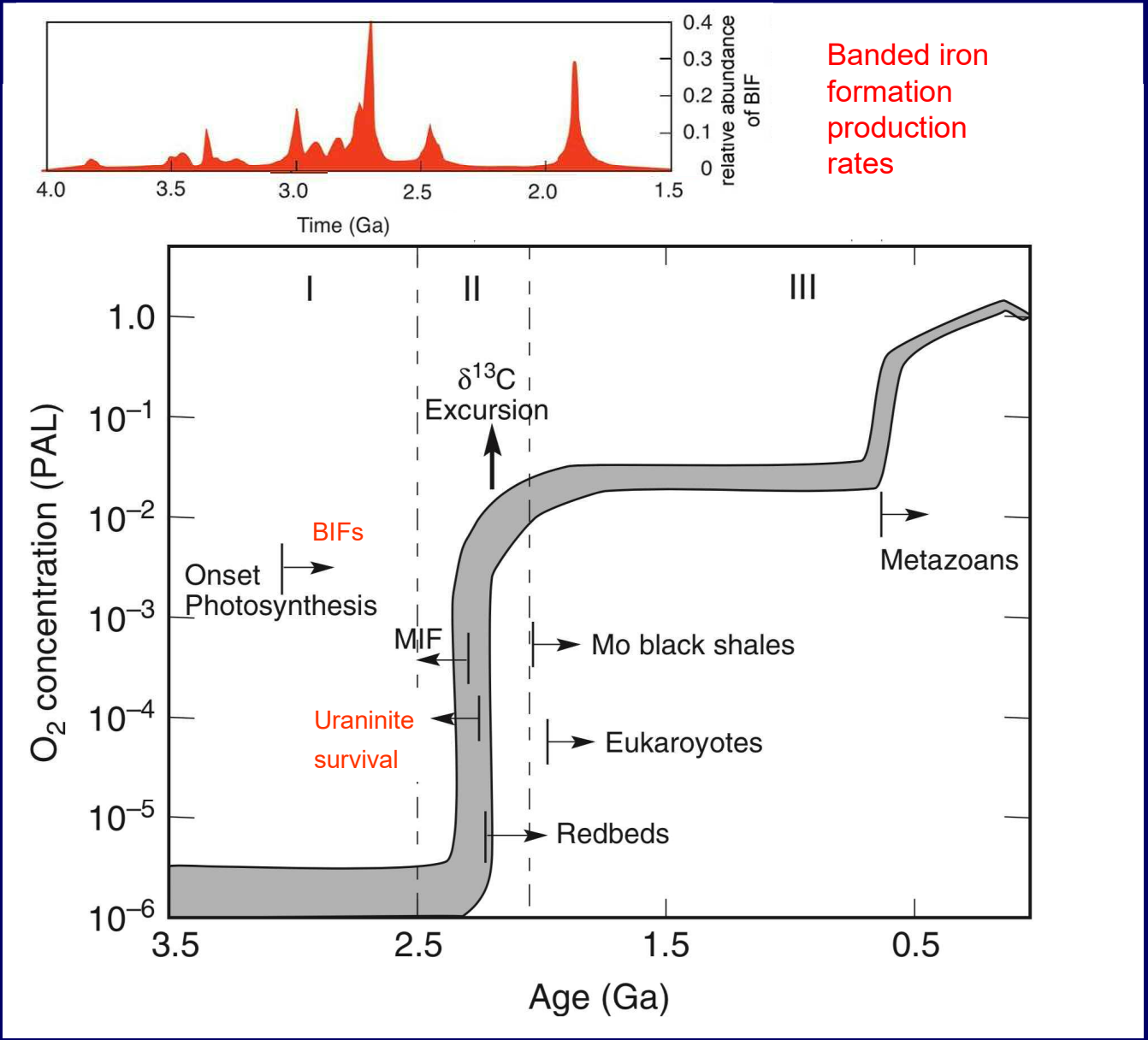
During the Proterozoic oxygen levels rose from <0.1% to >50% of current levels (PAL = present atmospheric level)

Comparison of the reducing stage with the modern oxygen cycle



Before the great oxidation event (GOE, 2.45-2.32 Ga) oxygen levels were < 0.1% of present levels

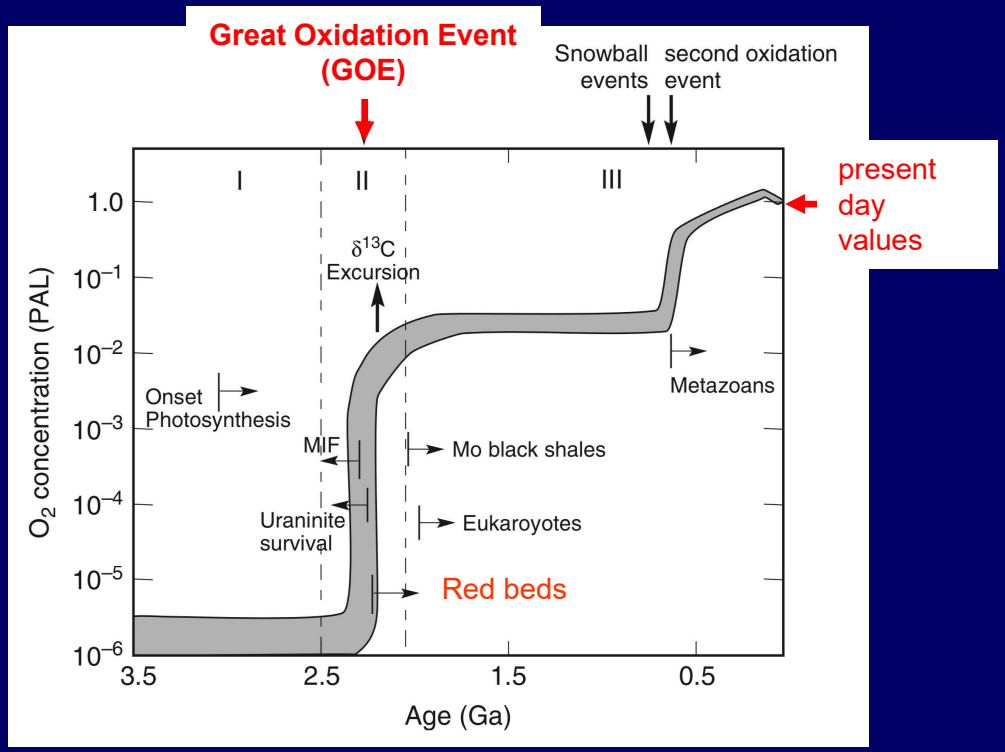
(I) Archaean Reducing stage: evidence



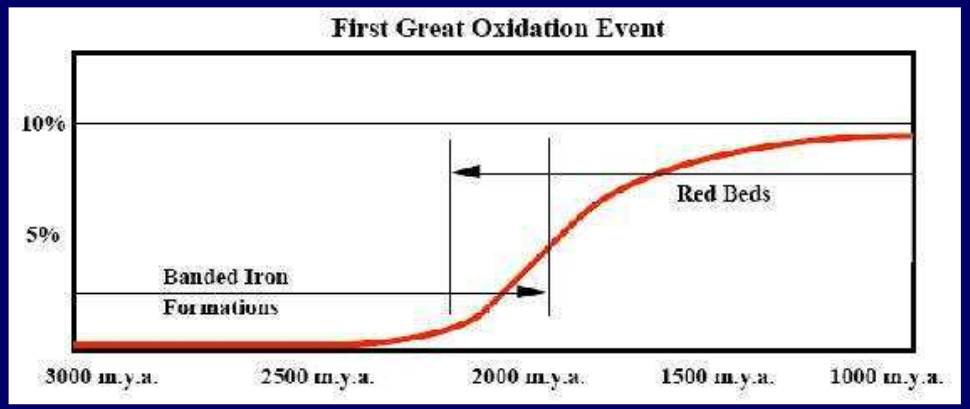
Distribution of BIF through time

+ redox sensitive detrital minerals in sedimentary rocks, notably uraninite (UO_2) and pyrite (FeS_2)

(II) Palaeoproterozoic Oxidizing stage: The GOE (2.45-2.32 Ga)

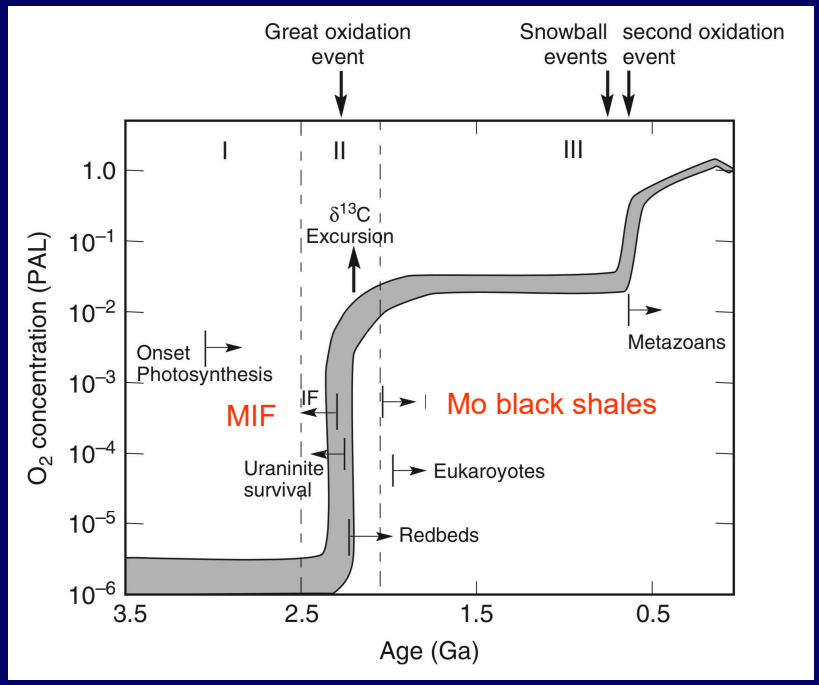


- *During the GOE atmospheric oxygen levels rose rapidly to 1-10% of present levels*
- *Red beds – detrital sedimentary rocks containing Fe^{3+} - oxide cements*
  - *Terrestrial origin (rivers, floodplains etc)*
  - *Weathering in presence of O_2 in atmosphere*



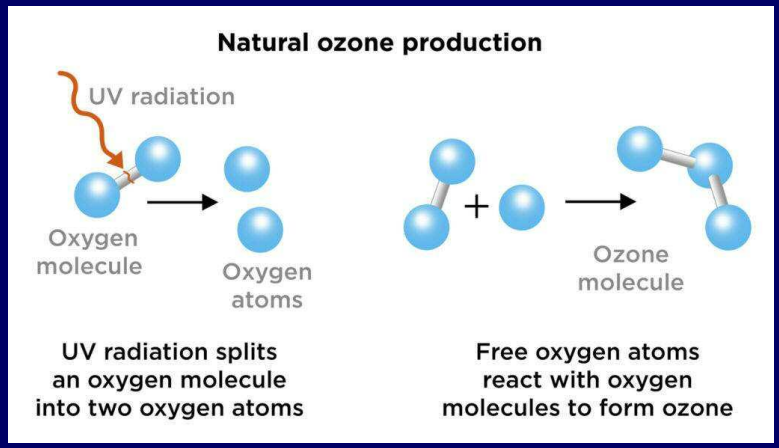
Red beds first appear ~2.3 Ga

Further evidence for the Great Oxidation Event



Other evidence for change in oxygen around 2.3 - 2.4 Ga:

- end of leaching of iron from paleosols
- end of major detrital uraninite deposition
- increase in abundance of evaporitic sulfate (including gypsiferous deposits in shallow marine settings)
- Molybdenum enrichment in black mudstone
- end to mass-independent sulfur isotope fractionation (MIF)



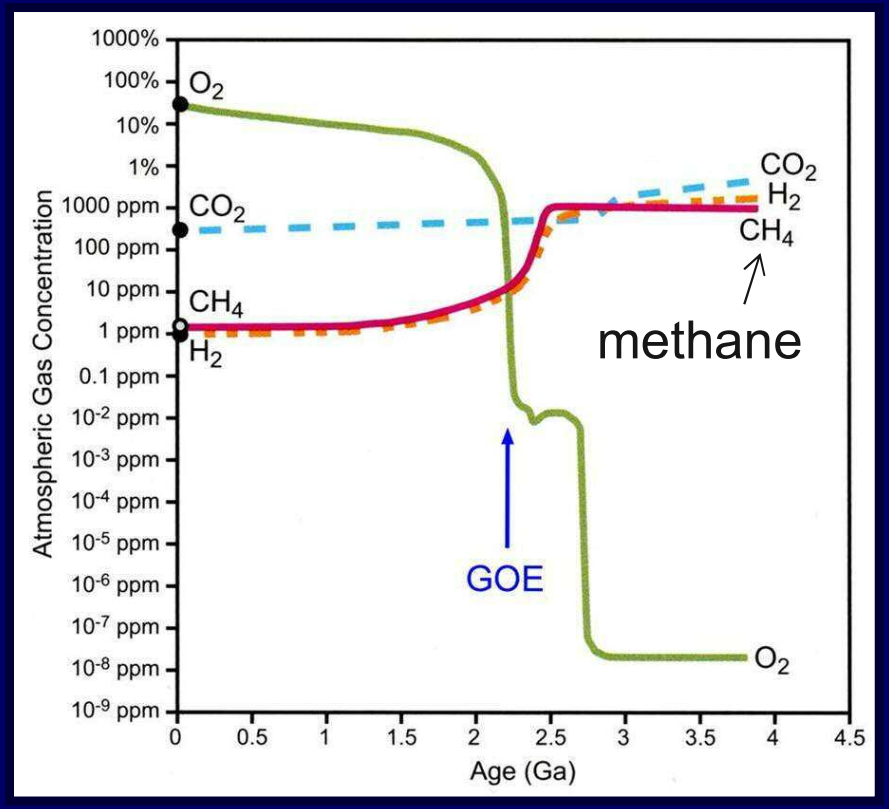
By 2316 ± 7 Ma an ozone layer had formed

What triggered the Palaeoproterozic rise in atmospheric oxygen?

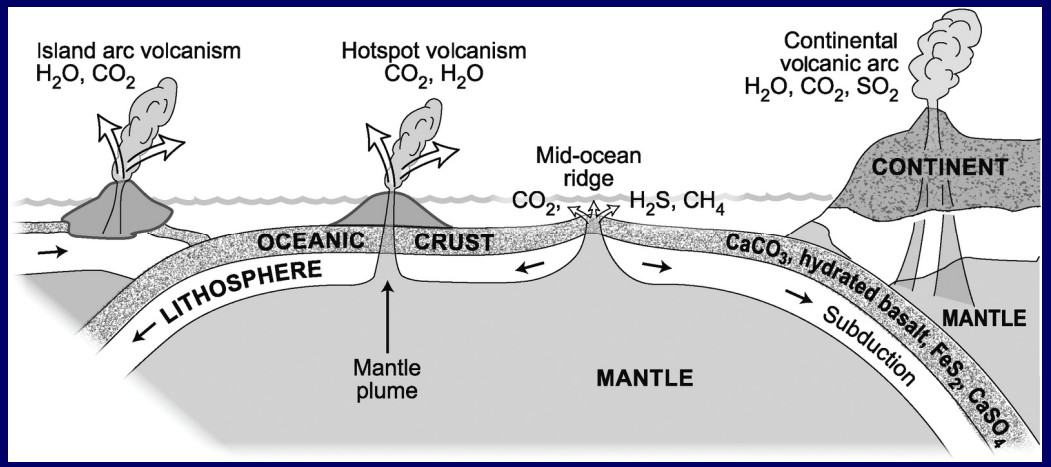
Change in amount of oxygen in the atmosphere per unit time = rate of production – rate of loss

- Amount of loss is dependent on how much oxygen is available, production is independent
- Increased rate of oxygen abundance driven by:
  - new source of oxygen
  - saturation of loss processes

(1) Oxygen as a waste product of photosynthesis by cynobacteria

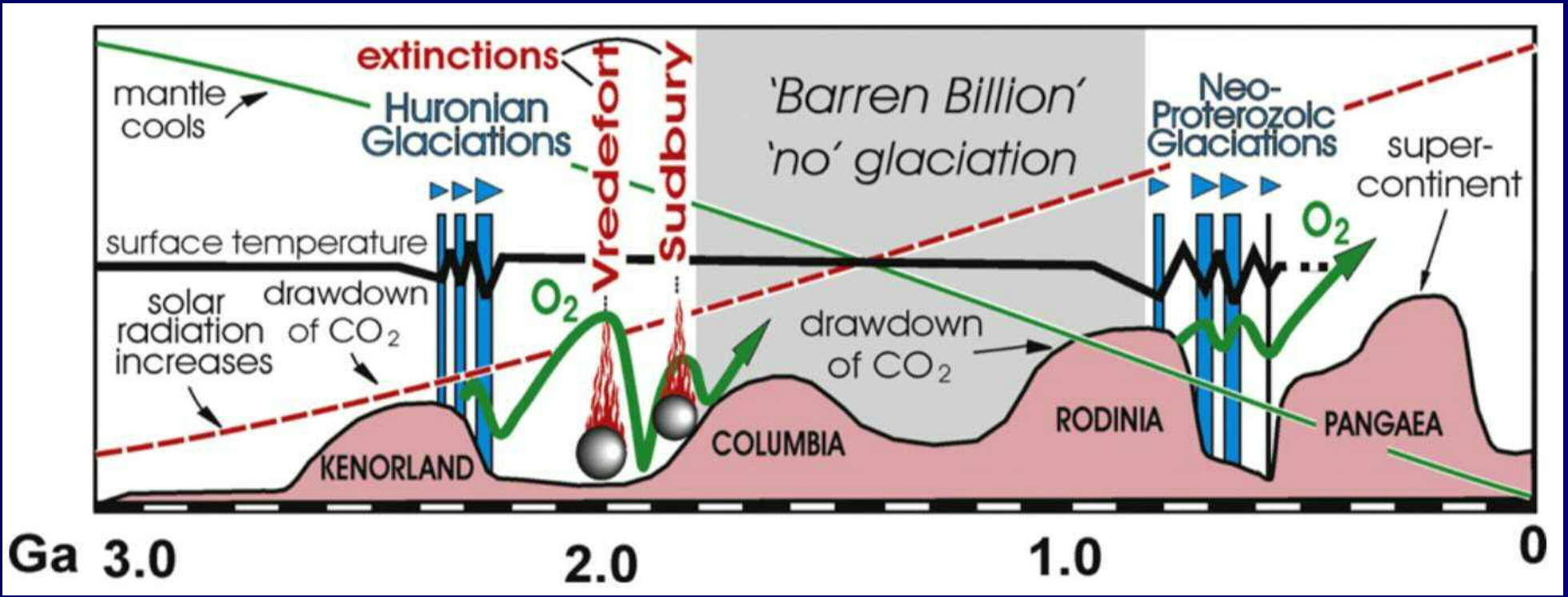


(3) Reduction of atmospheric methane



(2) Change in the redox state of volcanic gases, especially switch from H_2 to H_2O, due to shift to subaerial rather than submarine volcanism

Proterozoic glacial events



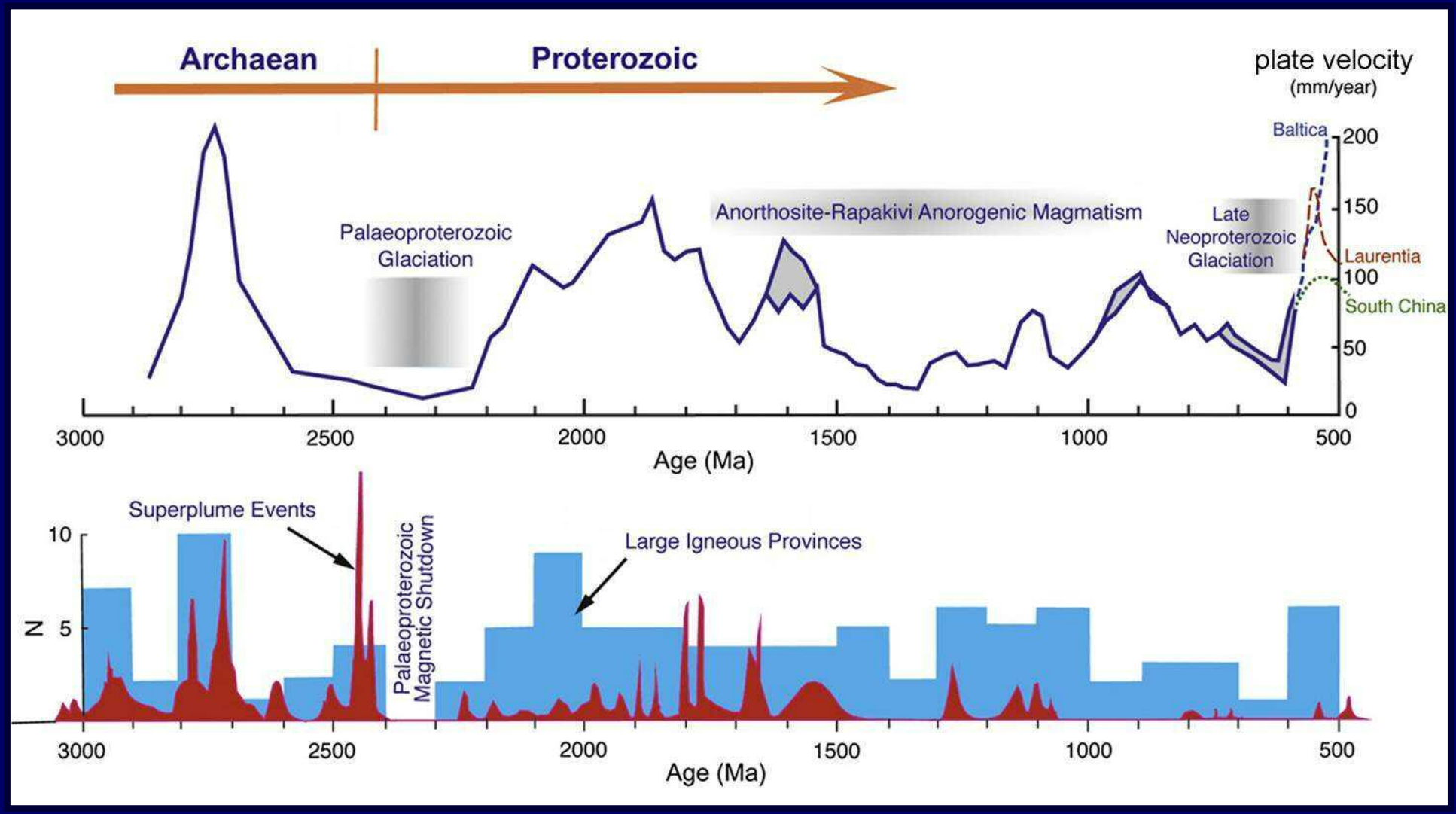
Surface temperature related to:

1. Solar luminosity = increase with Sun's age
2. Variation in albedo = distribution of land mass = plate tectonics
3. Variation in the greenhouse effect = volcanism = plate tectonics!

Palaeoproterozoic global glaciation (Huronian glacial event, 2420-2220 Ma):

Glaciogenic deposits indicate three glacial episodes in North America; a 4th is worldwide and includes glaciogenic deposits at equatorial palaeolatitudes in southern Africa, 2290-2250 Ma

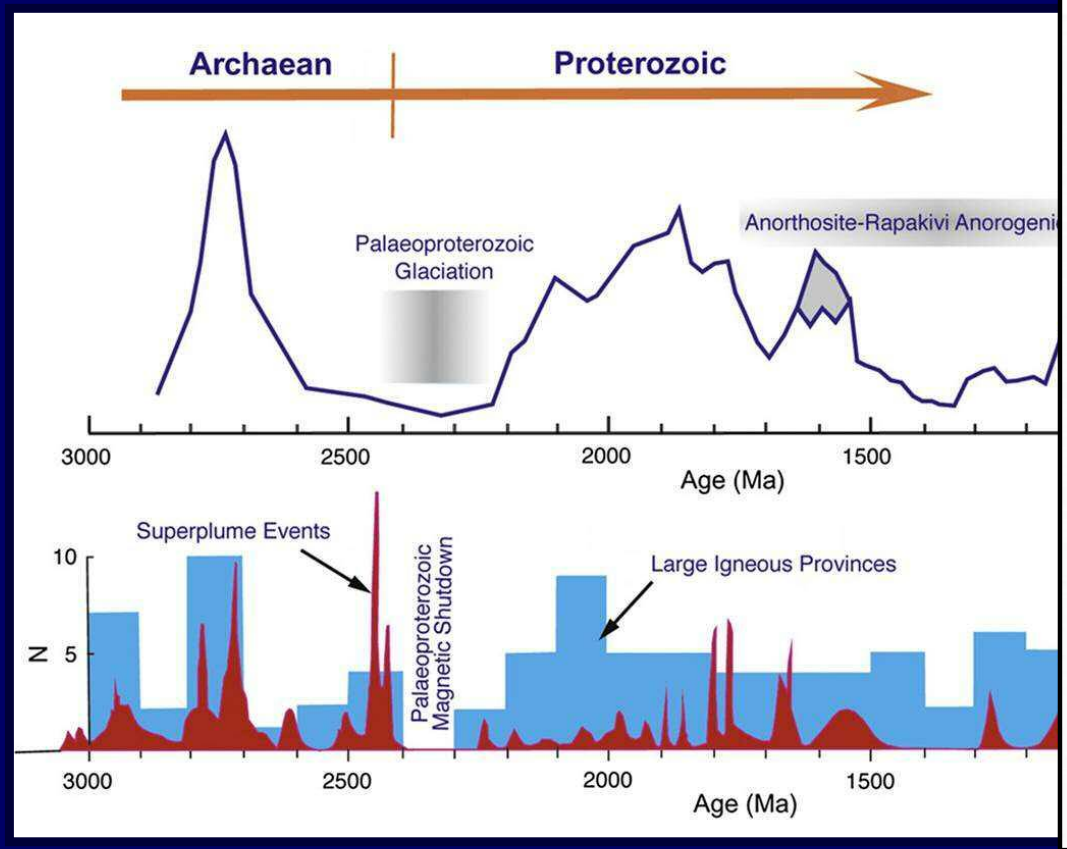
Causes of the Palaeoproterozoic “snowball Earth”

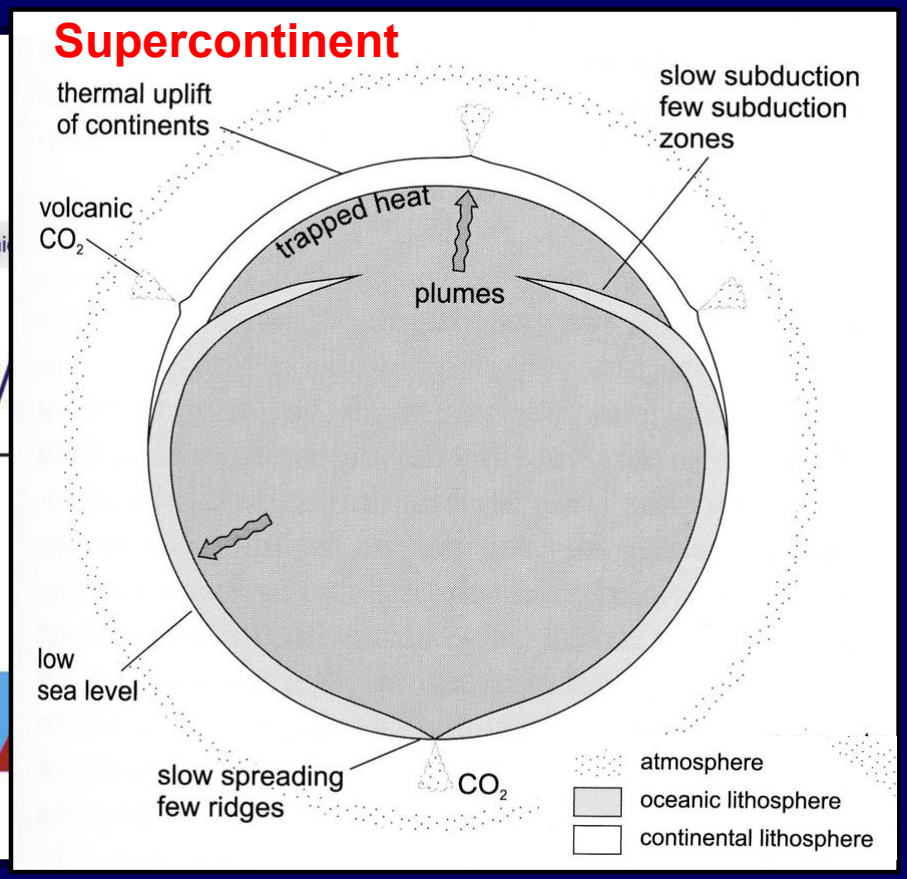


Superplume Event = short-lived mantle event (=100 Ma).

Many superplumes and smaller plumes impact base of lithosphere.

Causes of the Palaeoproterozoic “snowball Earth”





Slow moving continent → build up of heat under continent → thermal uplift of continent + low global magmatic activity on the sea floor → thermal subsidence (deepening) of oceans → net effect emergence of continents above sea level ↕ → weathering of supercraton + low magmatic activity → ↓atmospheric CO_2 → cooling

Increasing oxygenation of the atmosphere (GOE: photosynthesizing cyanobacteria) → removal of atmospheric CH_4 (strong greenhouse gas) → enhance cooling

Supercontinentality in context of weak solar radiation → Snowball Earth

Palaeoclimate indicators

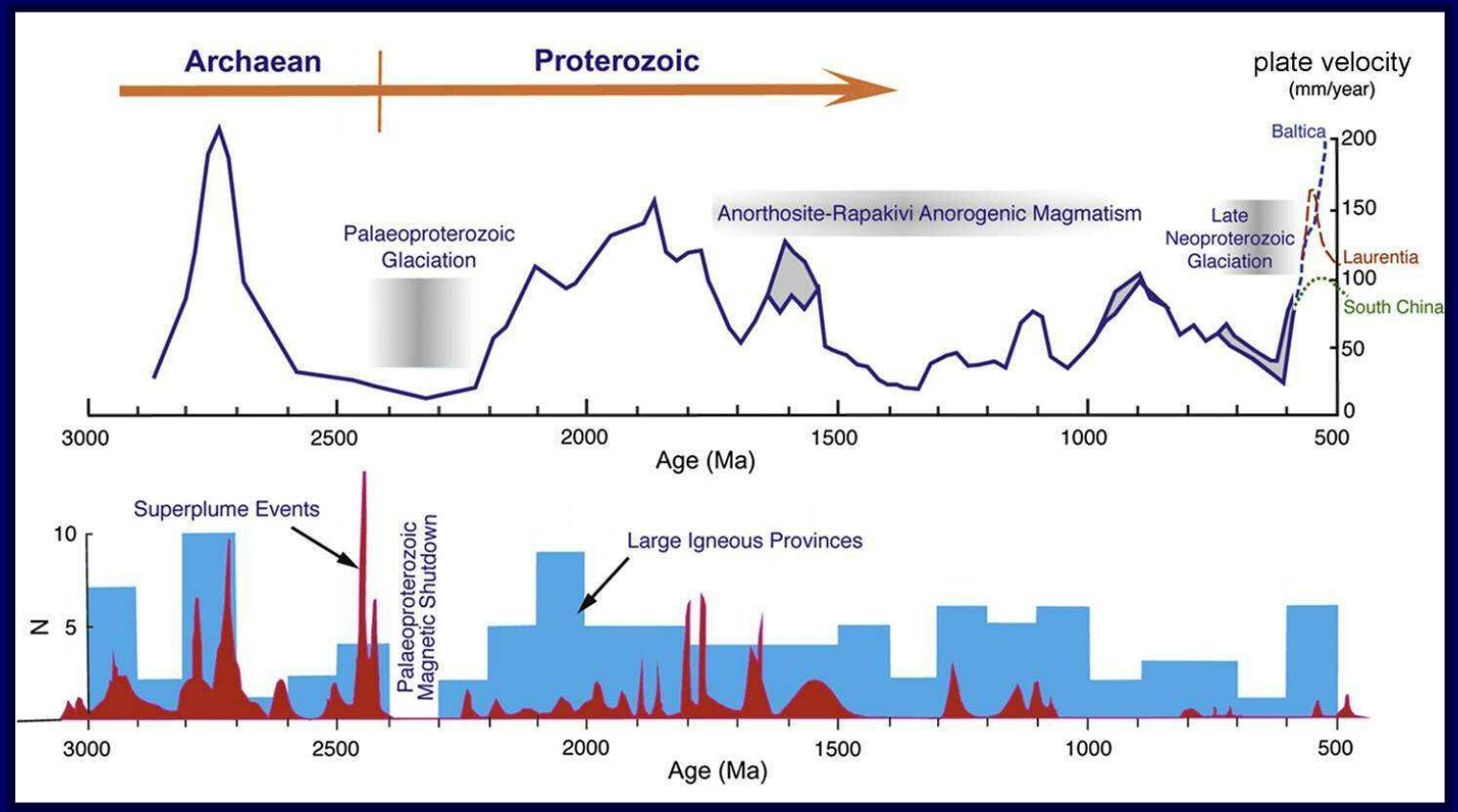
Diamictite (glacial tillite),
2.3 Ga Canadian Shield

Aeolian sandstones (Navajo sandstone, Utah, USA)

Indicator	Interpretation
Widespread glacial deposits	Continental glaciation; widespread cold climates
Coal	Abundant vegetation, moist climate, generally mid-latitudes
Eolian sandstones	Arid or semi-arid; generally but not always hot or warm temperatures
Evaporites	Arid or semi-arid; generally but not always hot or warm temperatures
Redbeds	Arid to subtropical; commonly semi-arid or arid
Laterites and bauxites	Tropical, hot and humid climates
Corals and related invertebrate fauna	Subtropical to tropical, nearshore marine

Plate tectonics and global warming from 2200 Ma

Palaeoproterozoic global glaciation ended with breakup of Kenorland



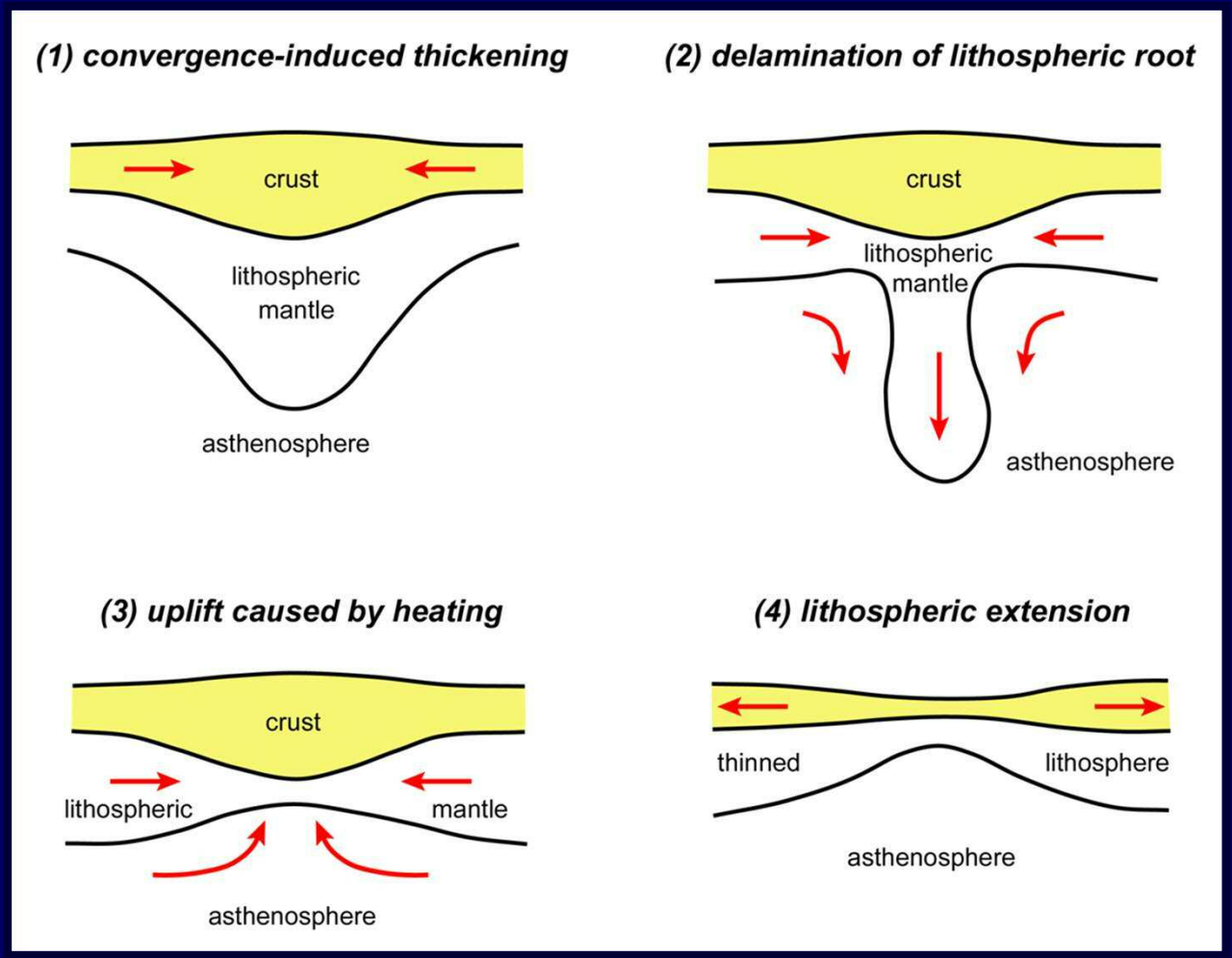
Onset of major rifting → renewed volcanic activity → ↑ atmospheric CO_2 → exit from glacial conditions

……and the continents dispersed

Continental breakup: The life and times of an orogenic belt

Thickening during collision of relatively cool, strong lithosphere

Heat can't escape and lithosphere heats up (weaker)



Delamination of the lithospheric root because it is denser than the underlying asthenosphere

Lithospheric mantle now thin

Intracontinental Rifting Indicated by:

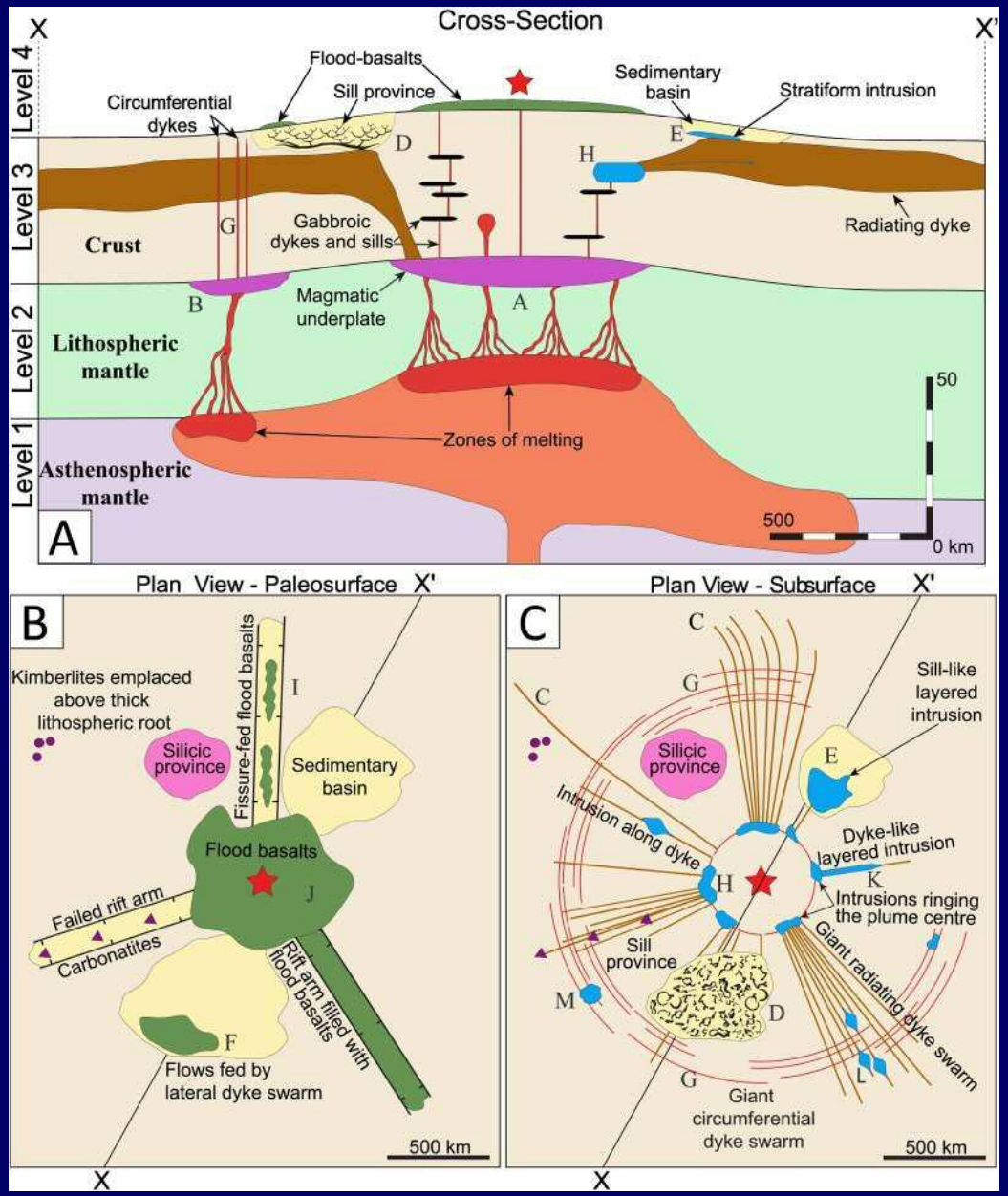
- dyke swarms
- rift basin fills

Uplift as warm asthenosphere replaces the cool lithospheric root

Crust heats up and becomes too weak to support its own weight → crustal extension and thinning

= intracontinental rifting and breakup of supercontinent

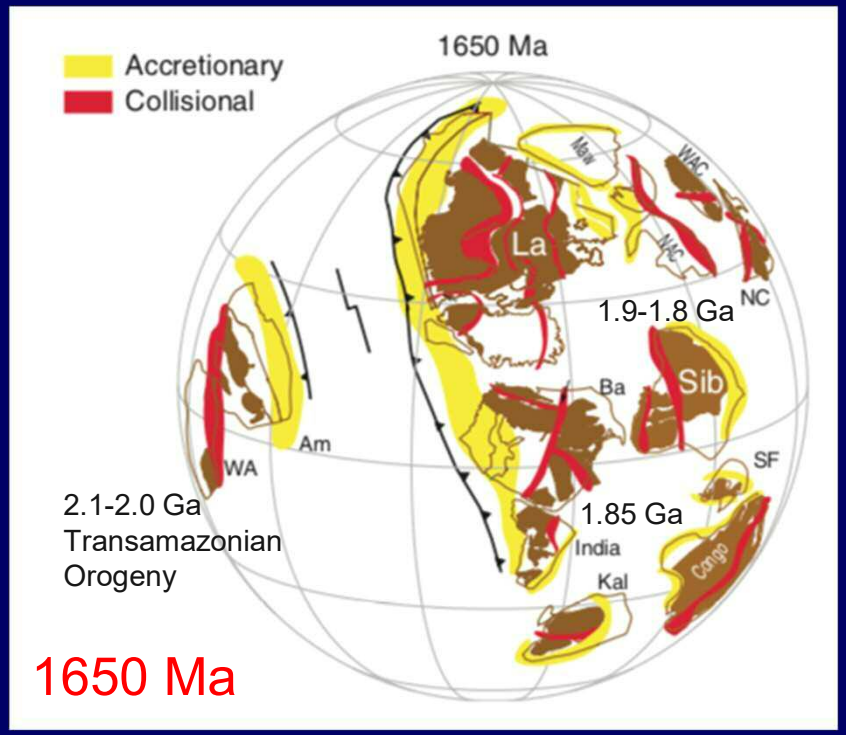
Palaeoproterozoic breakup of Kenorland after ~2200 Ma



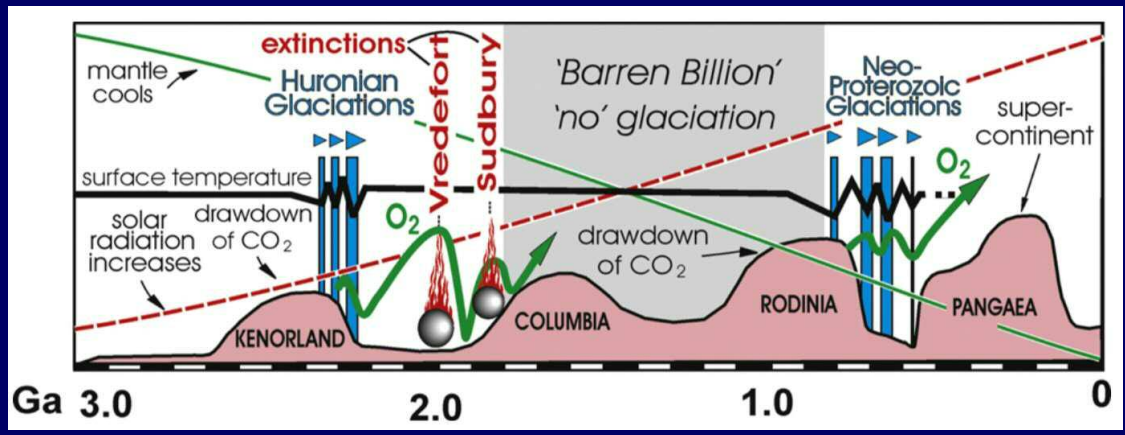
The plumbing of a large igneous province (LIP)

Dyke swarms have also been found on Venus and Mars

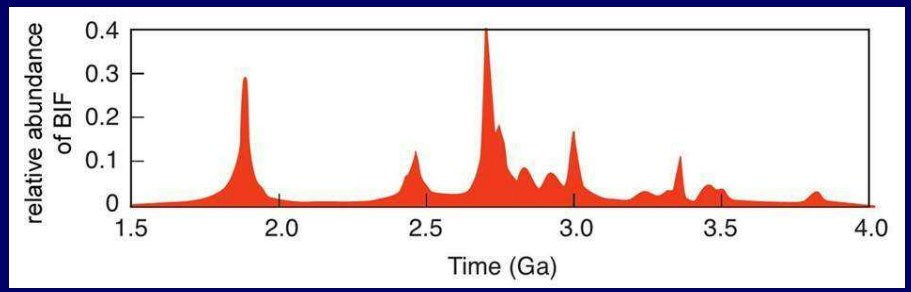
Mesoproterozoic: Earth's middle age (1780-850 Ma)



Global reconstruction of Columbia (Nuna) at 1650 Ma showing the distribution of collisional and accretionary orogens



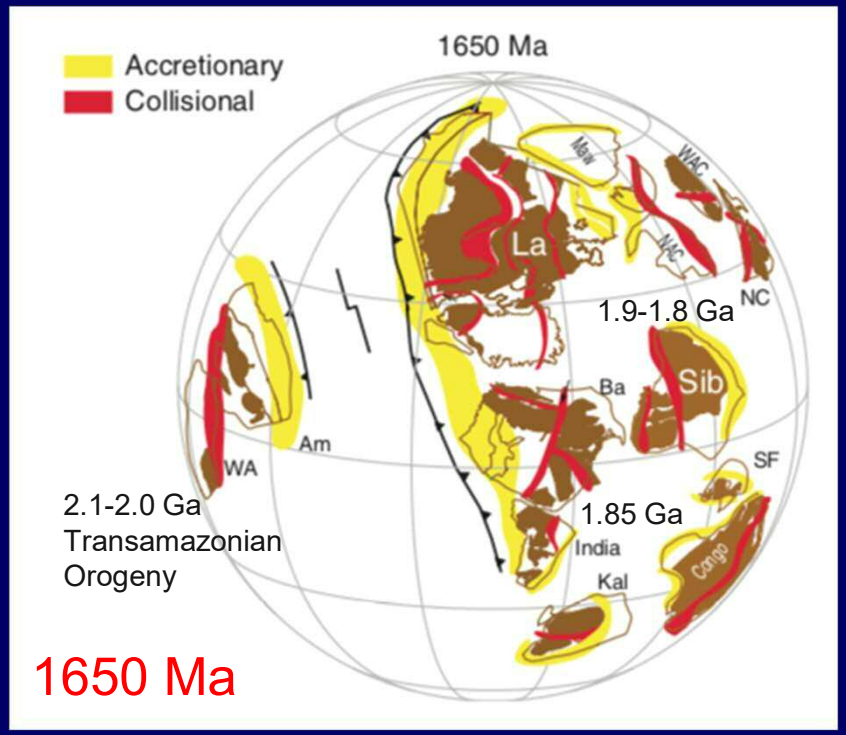
- From 2200 Ma: break-up of Late Archaean supercraton
- 2000-1800 Ma: initial assembly of Columbia



1800 Ma peak in BIF production

Large amounts of magmatic activity
→ ↑ atmospheric CO_2
→ oceans became anoxic again
→ peak in BIF production at 1800 Ma

The “boring billion”: Columbia and Rodinia



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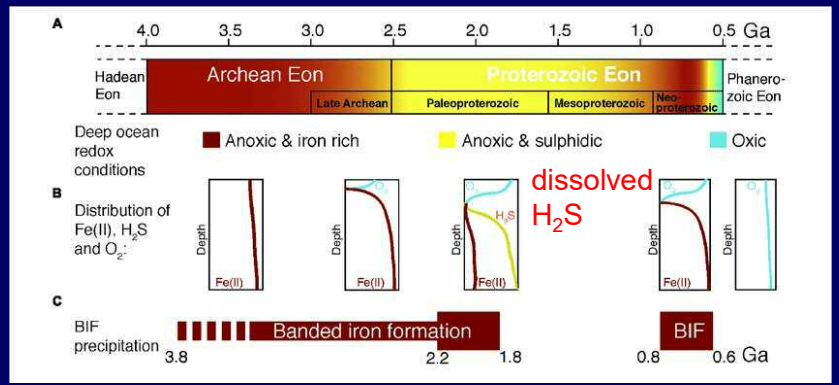
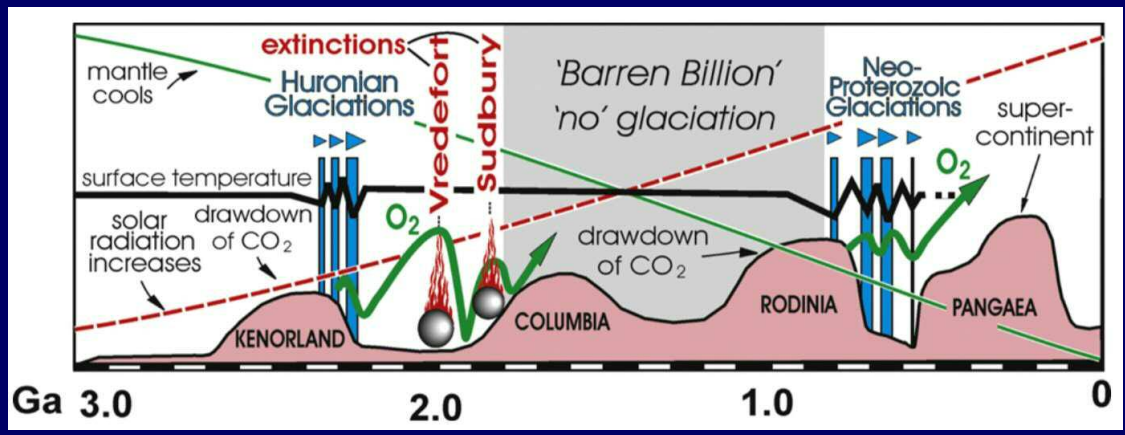


Sedimentary pyrite (FeS_2) deposition <1800 Ma



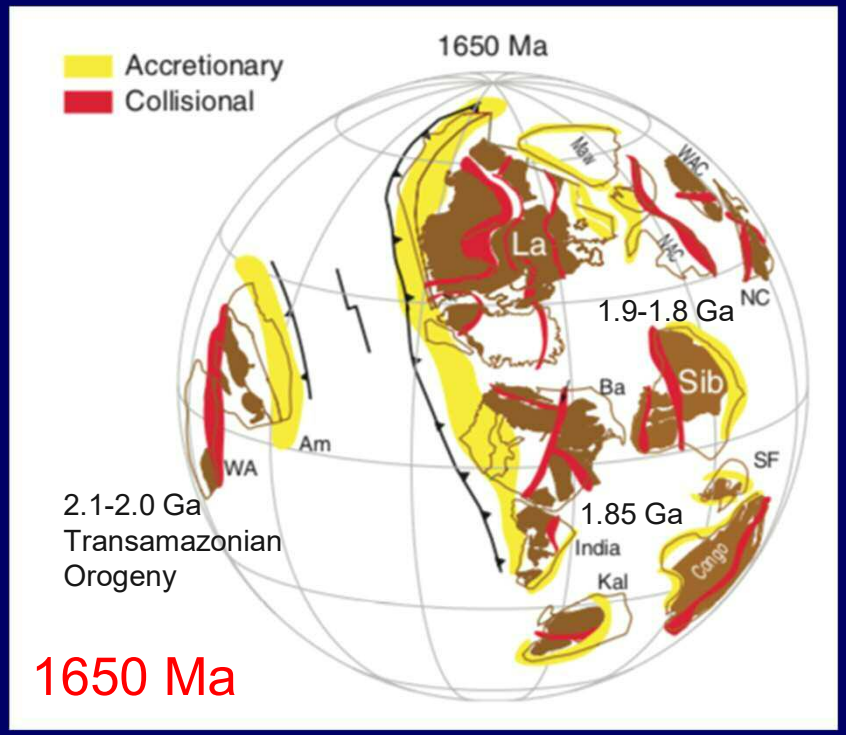
- 1800-1450 Ma: further accretion of continental terrains to periphery of Columbia

Relative environmental, evolutionary and lithospheric stability… but was it boring or an important watershed??

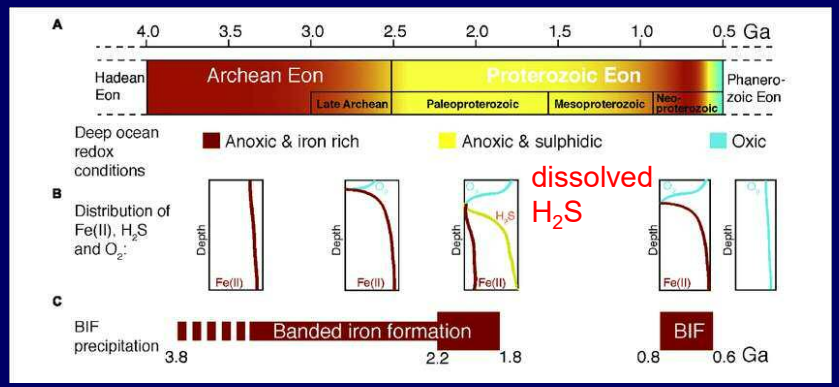
Supercontinent straddled equatorial and temporal climatic zone

No glacial deposits
→ climate and oceans relatively warm

The “boring billion”: Columbia and Rodinia



Global reconstruction of Columbia (Nuna) at 1650 Ma



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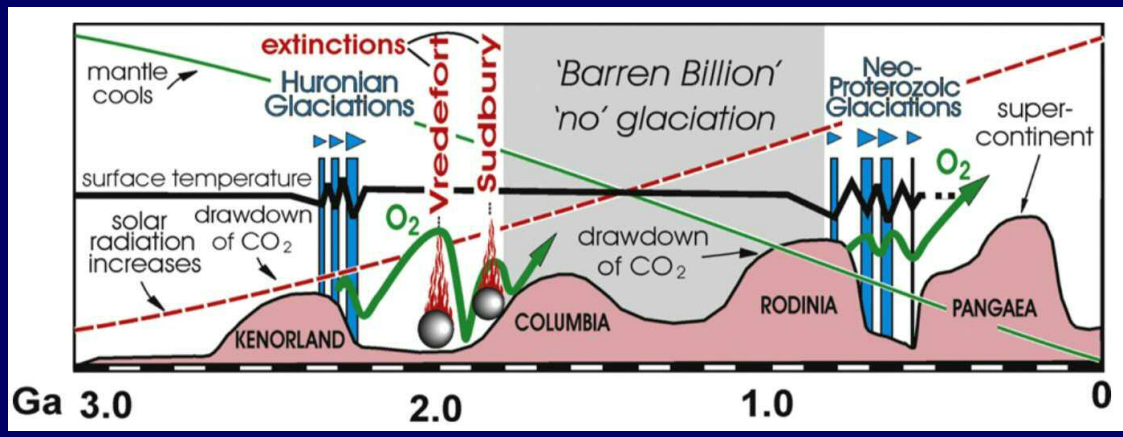
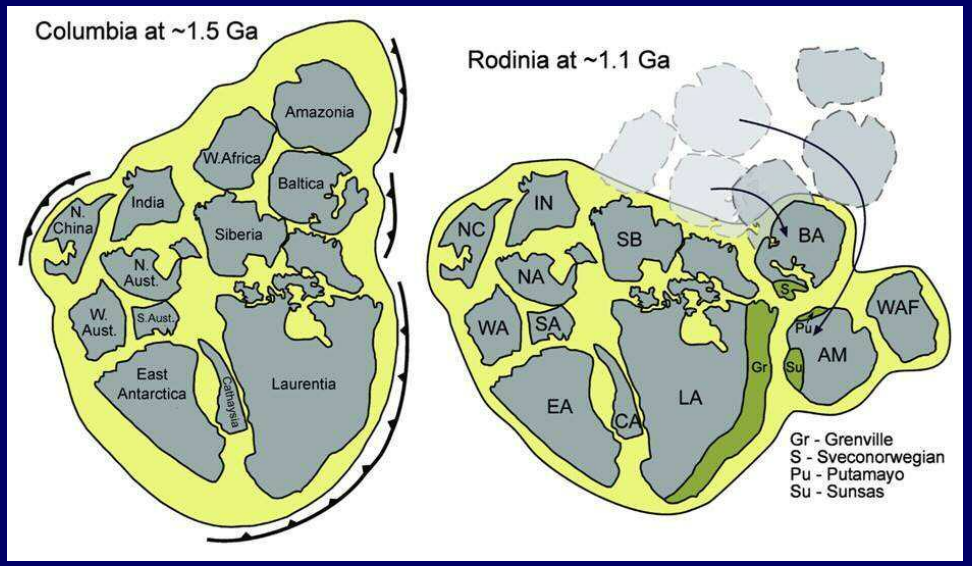


Oceans anoxic *but* no BIF production after 1800 Ma

Why?
Oceans anoxic and sulfidic = euxinic
S-rich from chemical weathering
→ soluble Fe^2 deposited as sedimentary pyrite (FeS2) instead of BIF

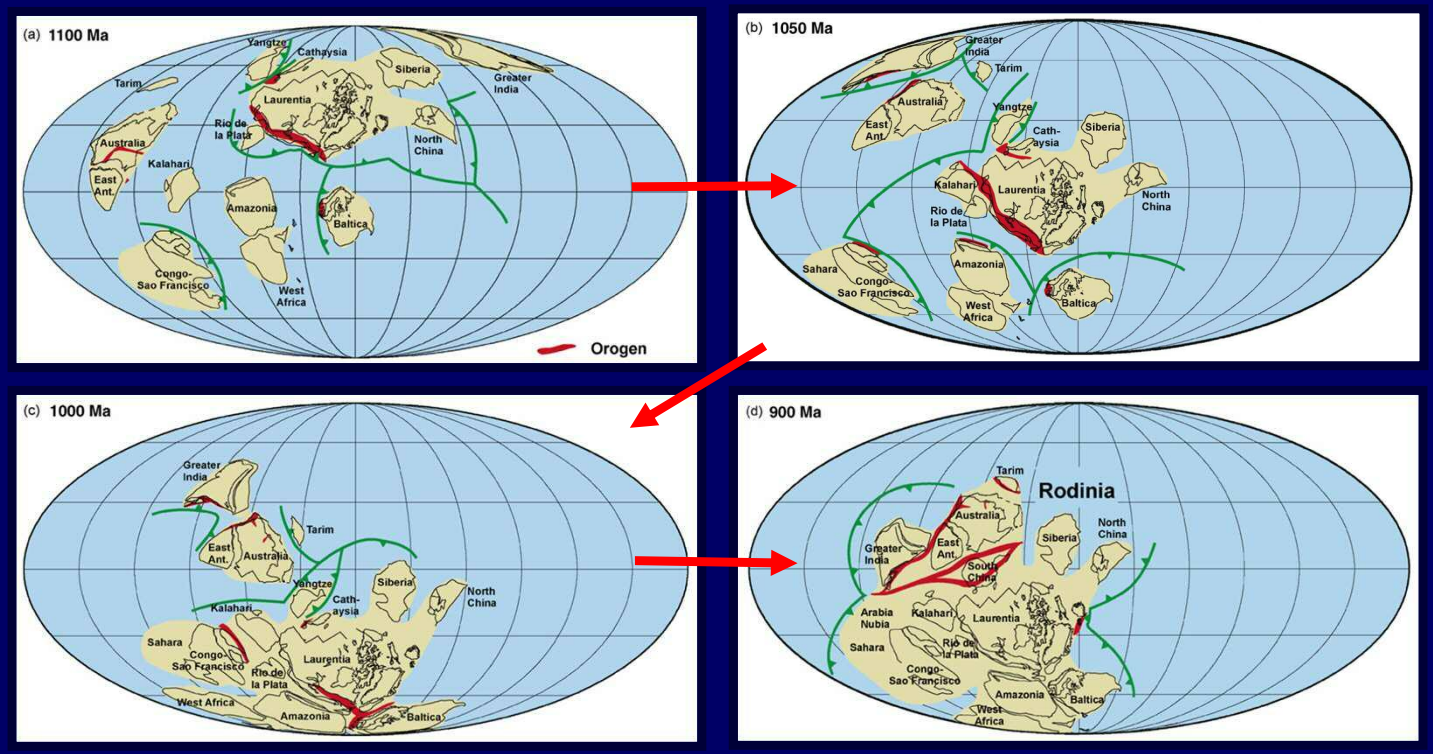
- 1400-1200 Ma: break-up of Columbia
- From ~1100 Ma: reassembly of continent to form Rodinia

Transition from Columbia to Rodinia

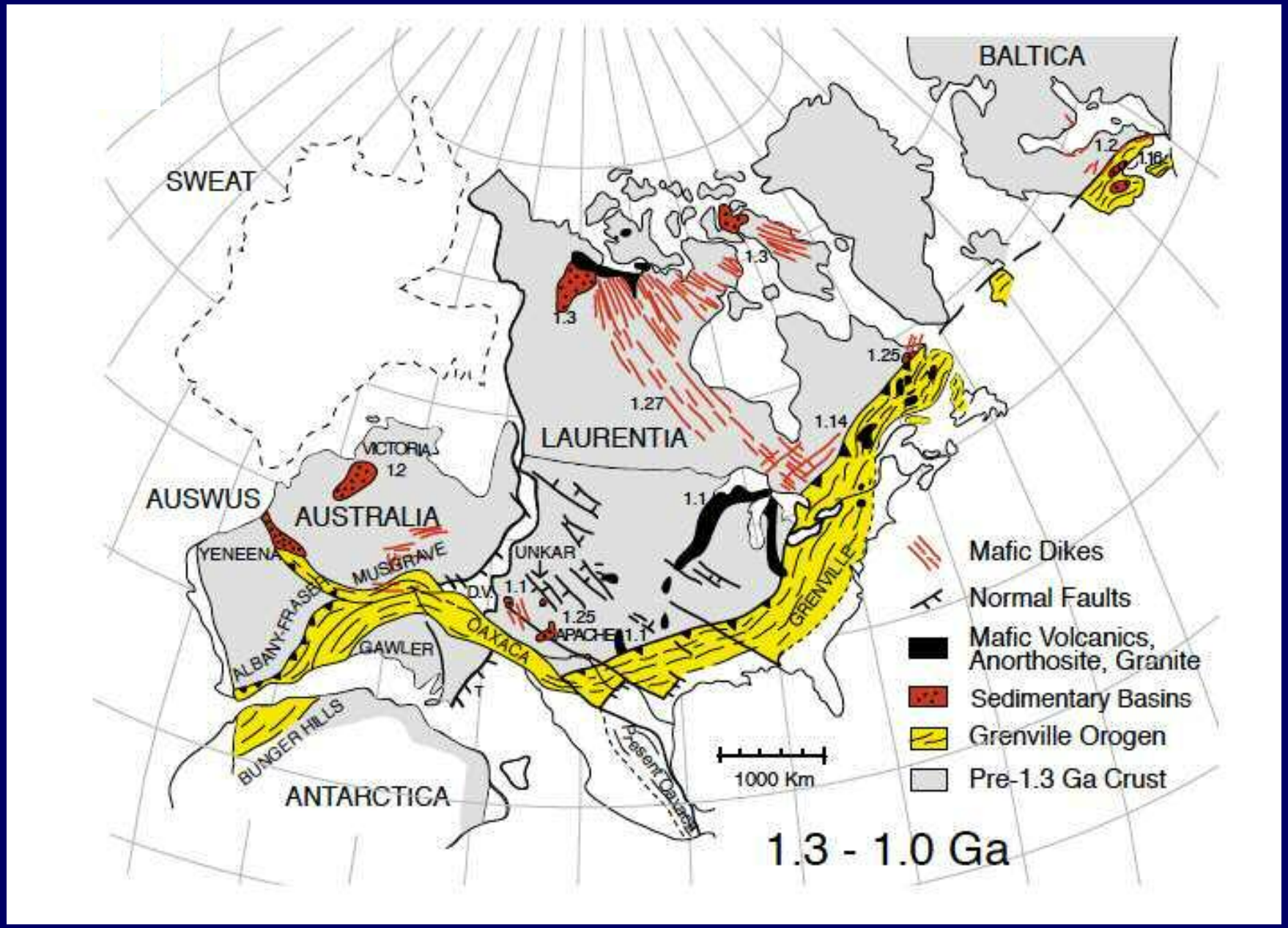


Reconfiguration without major fragmentation/dispersal/reassembly = long term tectonic stability

Alternative viewpoint: More dispersed distribution of continents prior to 1.1 Ga

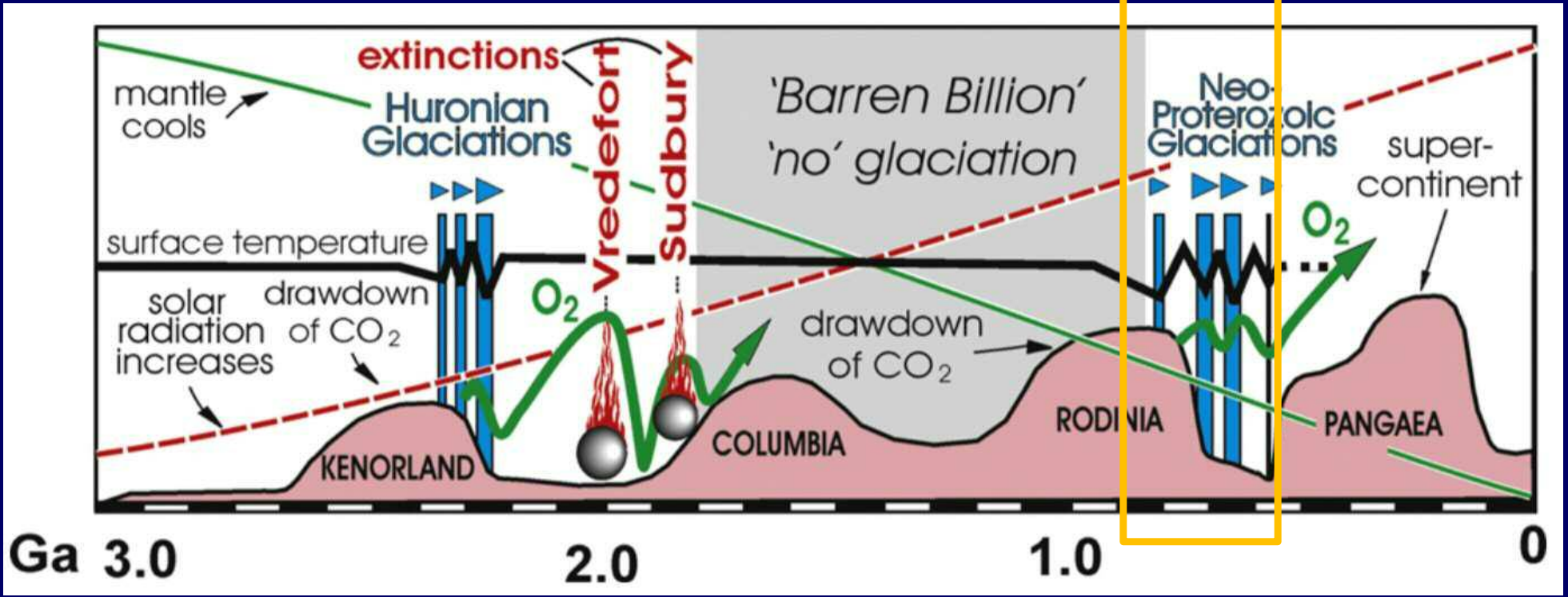


Grenville Orogeny (1085-985 Ma)



Neoproterozoic events (850 - 539 Ma)

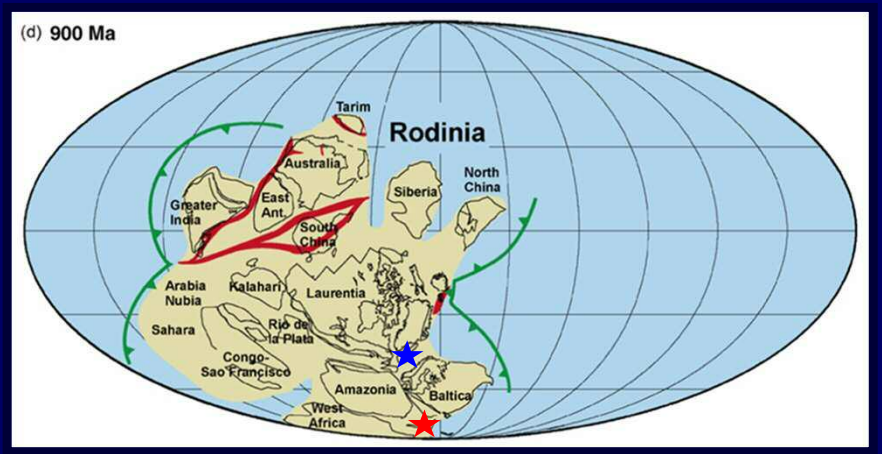
Neoproterozoic

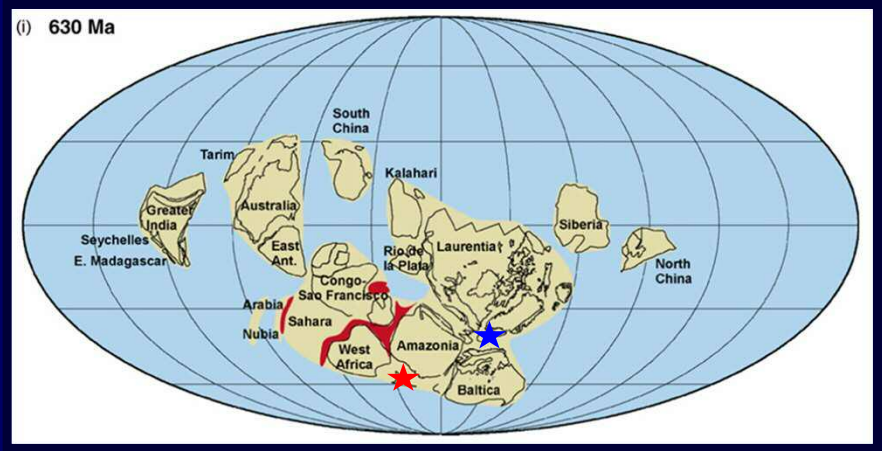


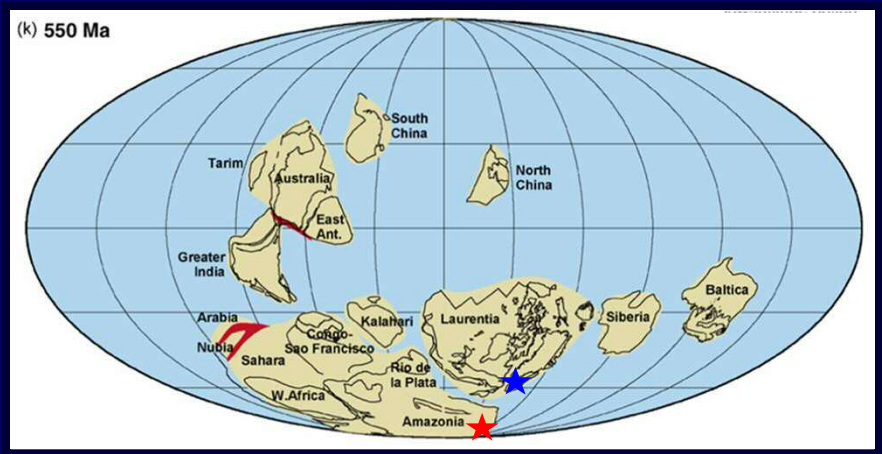
Key events

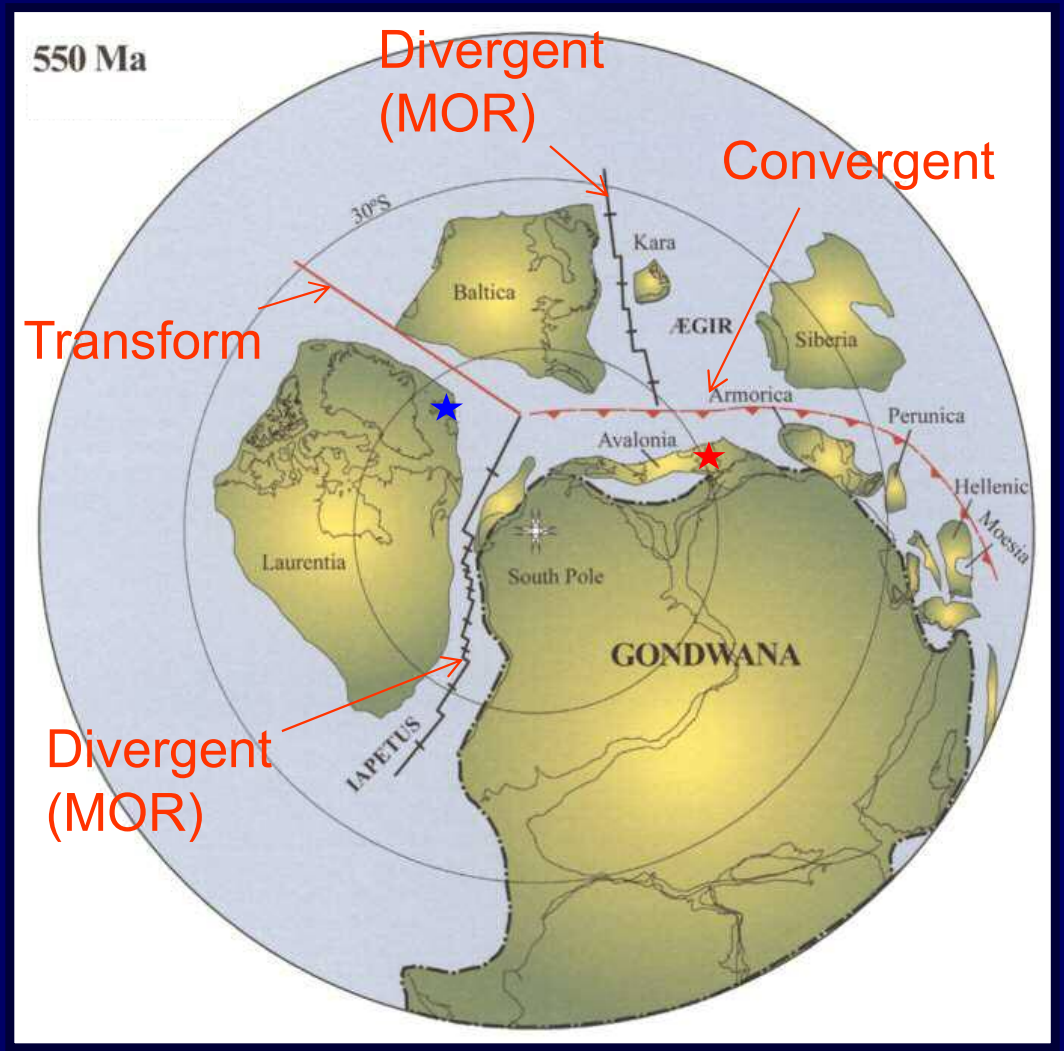
- break up of Rodinia
- onset of cold deep subduction
- second rise in oxygen
- Neoproterozoic glaciations

Break-up of Rodinia / formation of Iapetus ocean



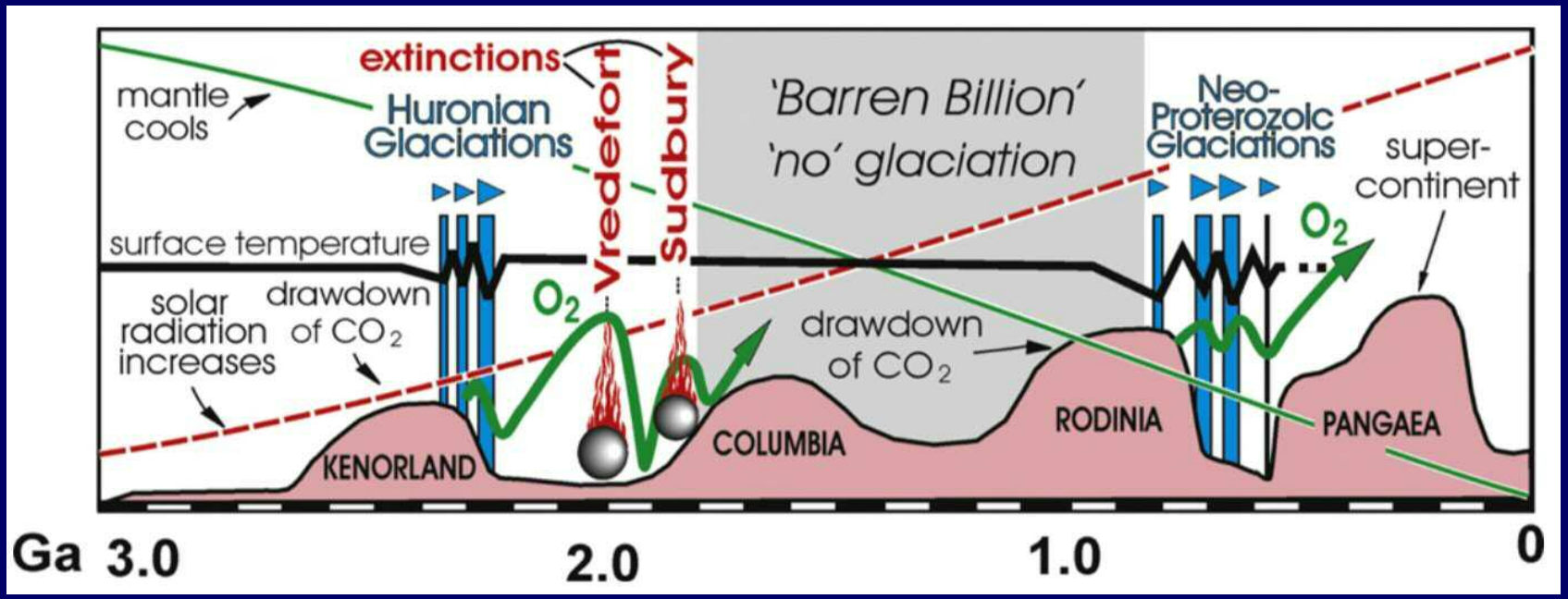




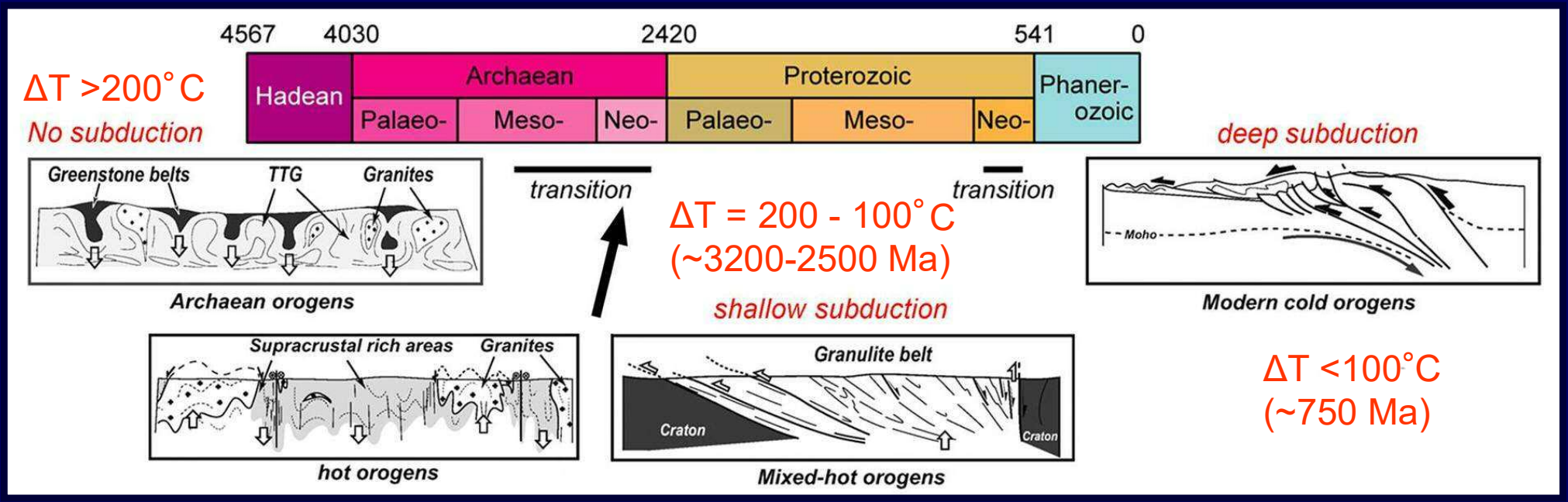


1. Formation of Rodinia, 1100 - 850 Ma
2. Break-up of Rodinia, 825 - 720 Ma
3. Formation of Pannotia, ~ 630 Ma
4. Break-up of Pannotia / Rodinia, 600 - 580 Ma

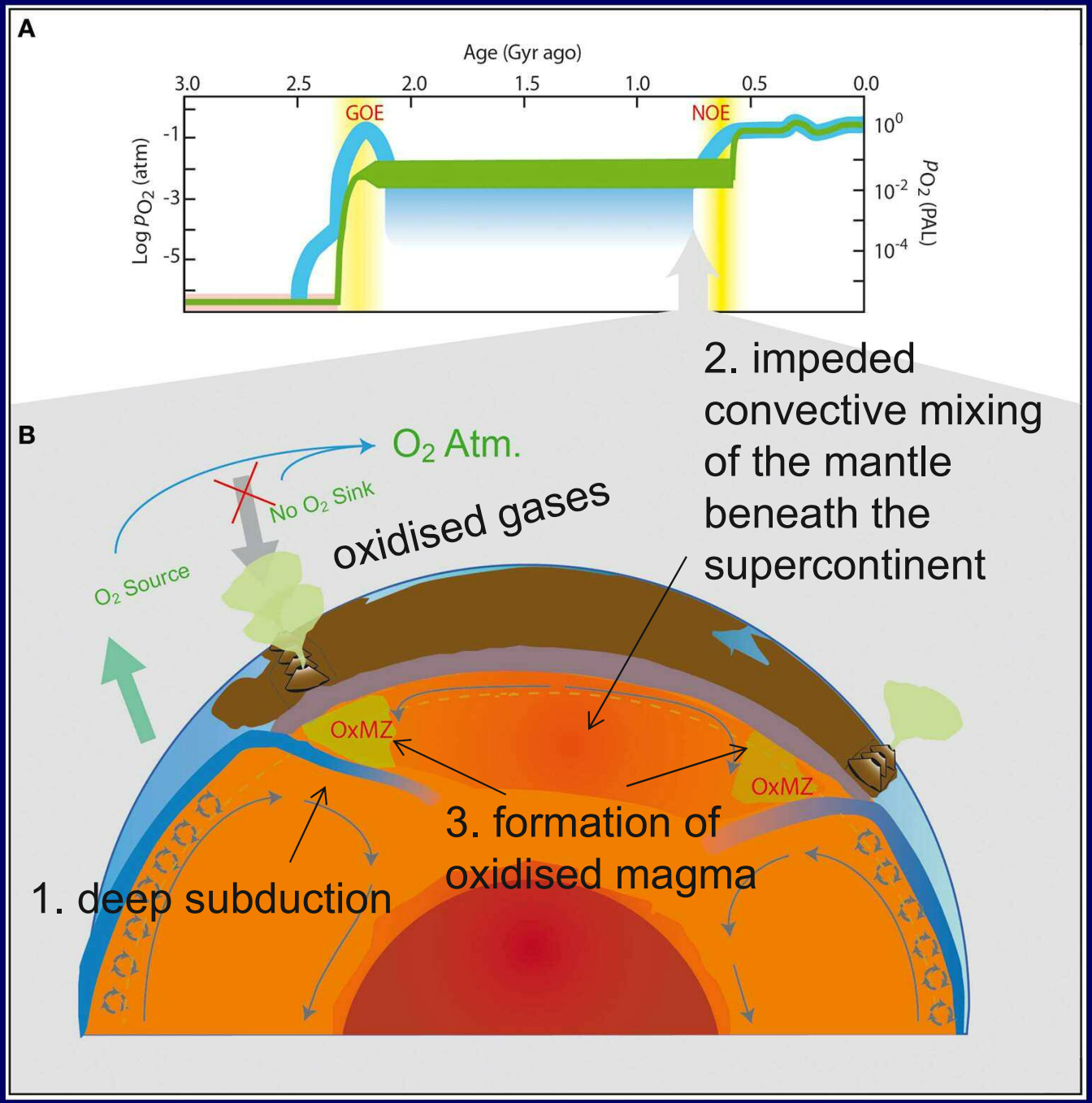
Evolution of plate tectonics in the Proterozoic



Since 3 Ga the mantle has progressively cooled



Neoproterozoic Oxygenation Event (800 – 650 Ma)



NOE = Neoproterozoic Oxygenation Event

- sharp rise in atmospheric oxygen content (from 10% present value to 50%+ or present values)

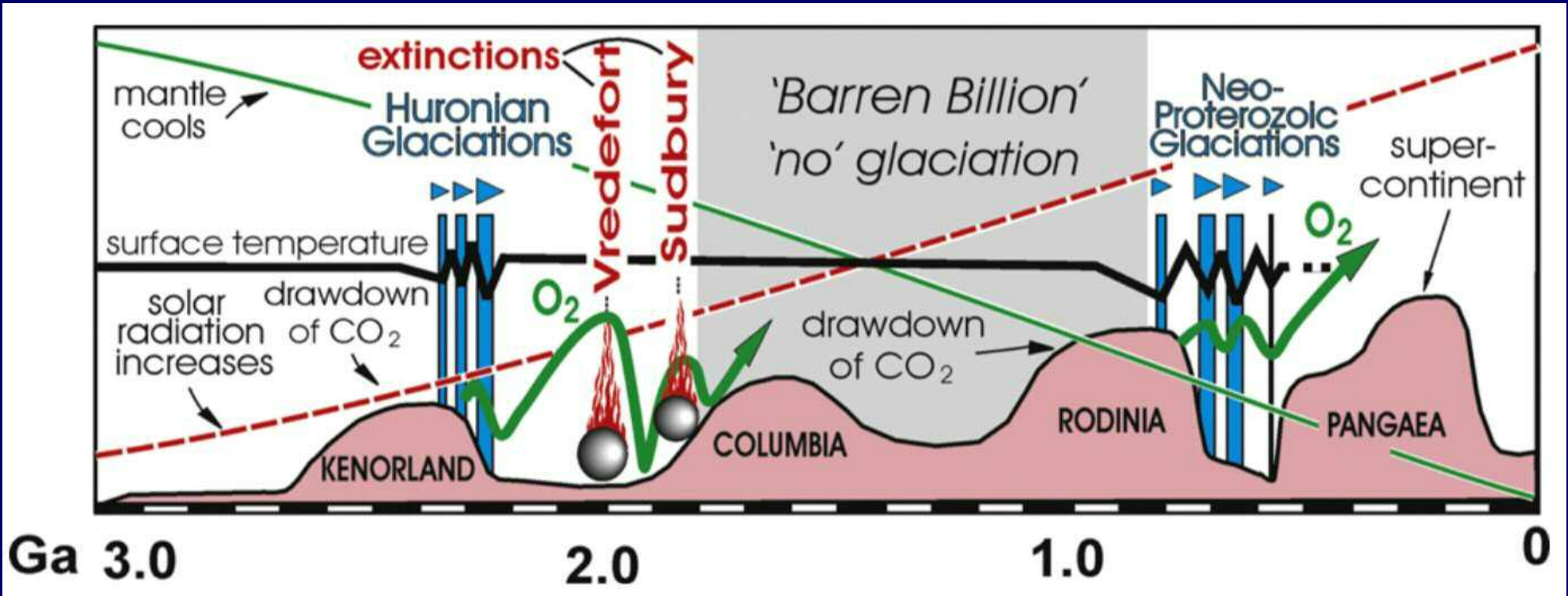
Possible causes:

- Change in tectonic style led to increase in atmospheric oxygen
- Uplifted mountains

→ enhanced weathering
→ nutrient runoff
→ stimulate photosynthesis
→ O_2 production & burial of carbon

The Neoproterozoic Snowball Earth (720 - 635 Ma)

As the supercontinent Rodinia was in the process of breaking up, a second series of extended glacial periods occurred



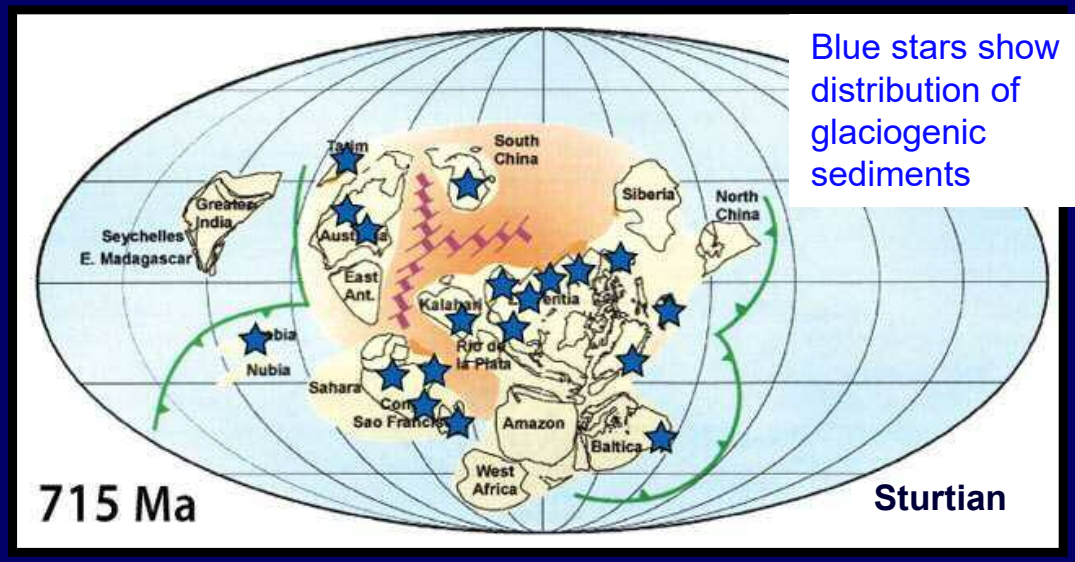
Extended periods when ice sheets formed at sea level down to equatorial latitudes ("Snowball Earth").

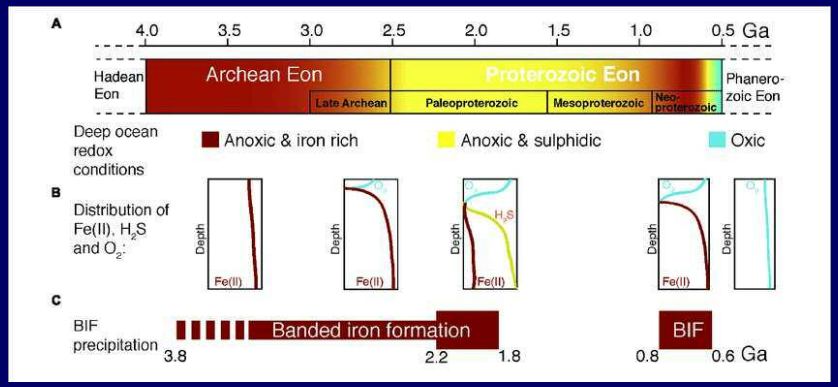
Glaciations lasted 10's Ma followed by hot periods (cold-hot cycles)

Paved way for Cambrian explosion of life

Evidence for the Snowball ice sheets

Evidence for snowball ice sheets comes in part from the distribution of glaciogenic deposits and BIFs





The Neoproterozoic Snowball Earth cycle

Trigger: high standing continents all in tropical latitudes (warm, humid) → enhanced chemical weathering → enhanced CO_2 draw down from atmosphere → cooling

+

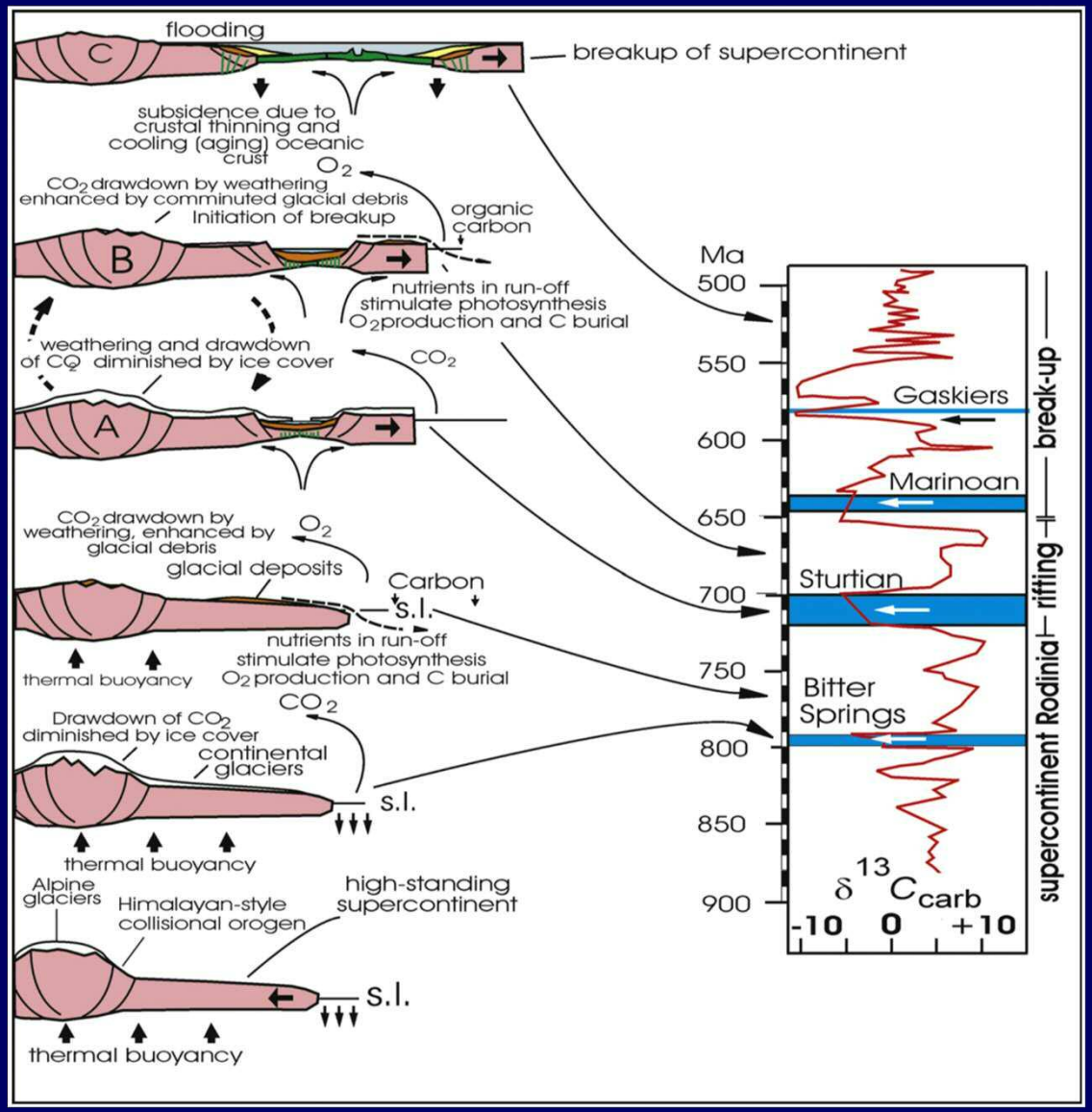
Enhanced weathering → nutrient runoff → stimulate photosynthesis → O_2 production and C burial

Albedo effect of snow / ice cover enhances cooling

Ice cover prevents CO_2 draw down (no weathering, carbonate burial, nor photosynthesis); volcanic CO_2 builds up

Greenhouse warming melts ice faster than renewed CO_2 draw down cools it → hot-house Earth (cap carbonates)

Extreme weathering (acid rain) draws down CO_2 and the hothouse is cooled



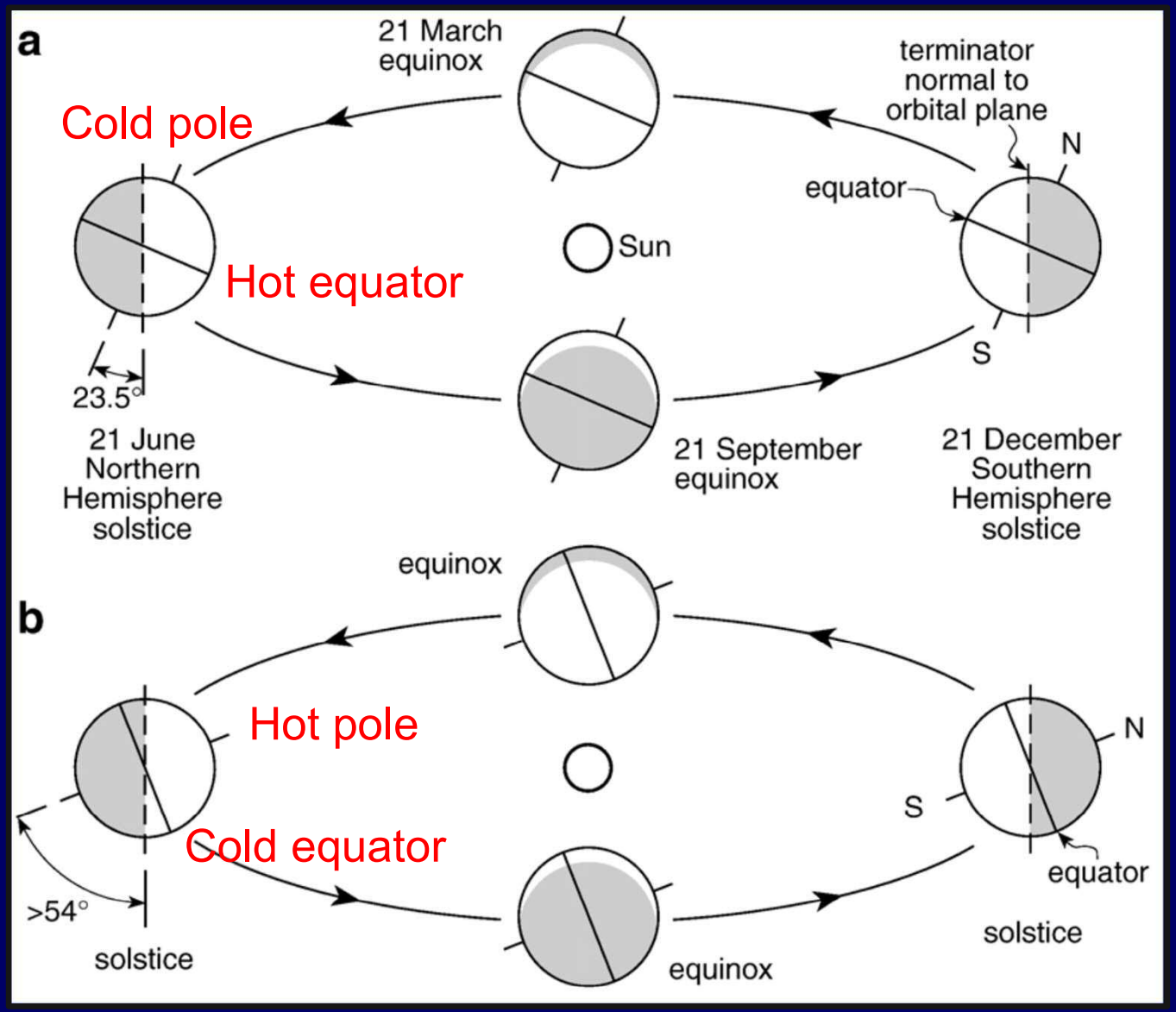
2-3 Snowball events from 800 - 575 Ma

Alternative hypotheses have been proposed

Alternative hypotheses to the Snowball Earth

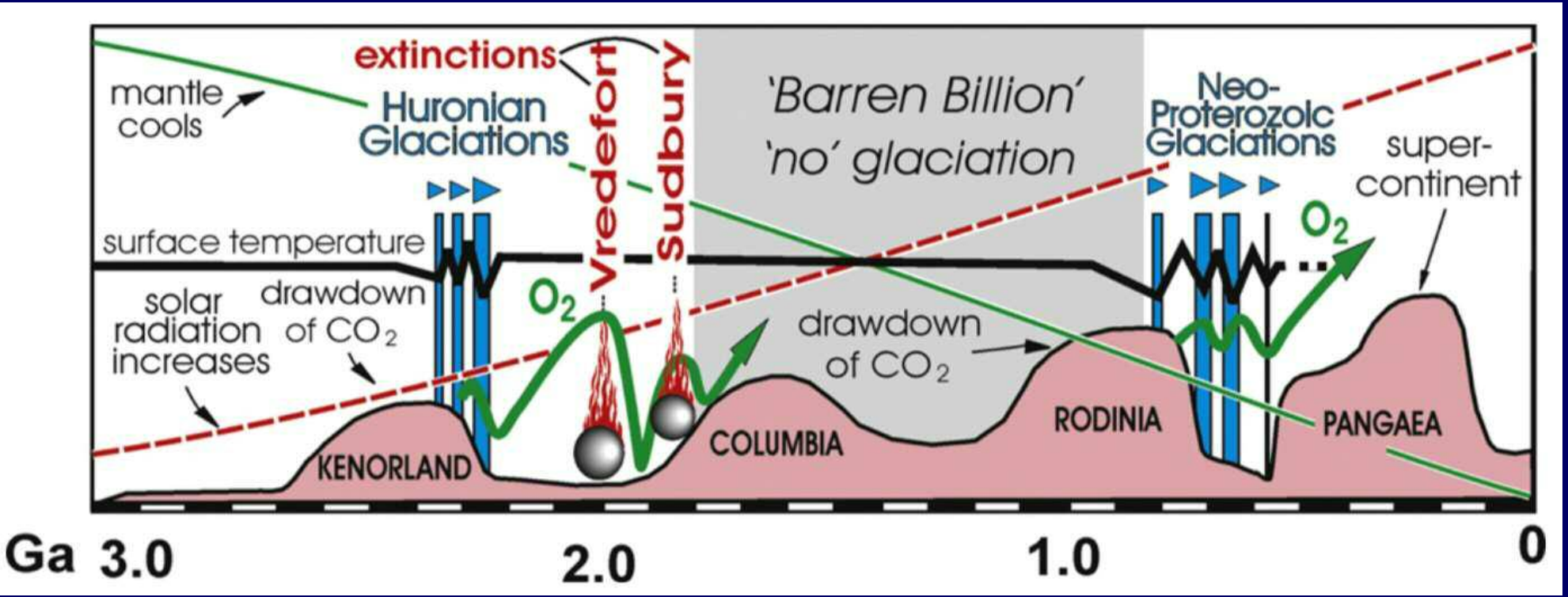
Earth's axis is tilted at 23½^0 to a line drawn perpendicular to the plane of its orbit

If the angle of obliquity was >54^0



Supports strong observed seasonality in Neoproterozoic low latitude deposits but no obvious mechanism for change in obliquity by Ordovician

Palaeoproterozoic versus Neoproterozoic Glaciations



- The Palaeoproterozoic Huronian glaciations are thought to have resulted mainly from the combined effects of supercontinentality and weak solar radiation
- Neoproterozoic glaciations occurred ~1.4 Ga later when solar radiation was greater, lower CO_2 content of the atmosphere (drawdown from weathering and mantle cooling), and the location of the fragmenting supercontinent Rodinia at the equator, are thought to be the main causes
- Amalgamation of the supercontinent Columbia in the 'Barren Billion' shows little evidence of significant climate change, and suggests there was a balance maintained between weathering intensity, solar luminosity and mantle temperatures that did not lead to dramatic changes in climate or biological productivity

Summary of key events in the Proterozoic